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EMU Filamentalist Assembly

EMU - The Expandable Multi-material Unit

VERSION 0.9 BETA - 2026-08-12

WHAT YOU'LL NEED

- Flush cutters
- Soldering iron
- Solder
- Wire stripper
- Crimping tool
- 300mm Ruler
- Measuring tape
- 2mm & 2.5mm Ball End Hex key
- Countersinking tool or 4mm drill bit to chamfer the PTFE tubes
- PTFE cutter tool (optional)
- A kitchen blowtorch or heat gun (optional but highly recommended)

2

To have the best chance of success, the following print settings are highly recommended.

3D PRINTING PROCESS
Fused Deposition Modeling (FDM)

MATERIAL
ABS/ASA. Glass and CF filled material are not recommended

LAYER HEIGHT
Mandatory: **0.2mm, including first layer!**

EXTRUSION WIDTH
0.4mm, Arachne slicing engine.
Infill can be printed wider (0.45/0.48) to increase layer bonding.

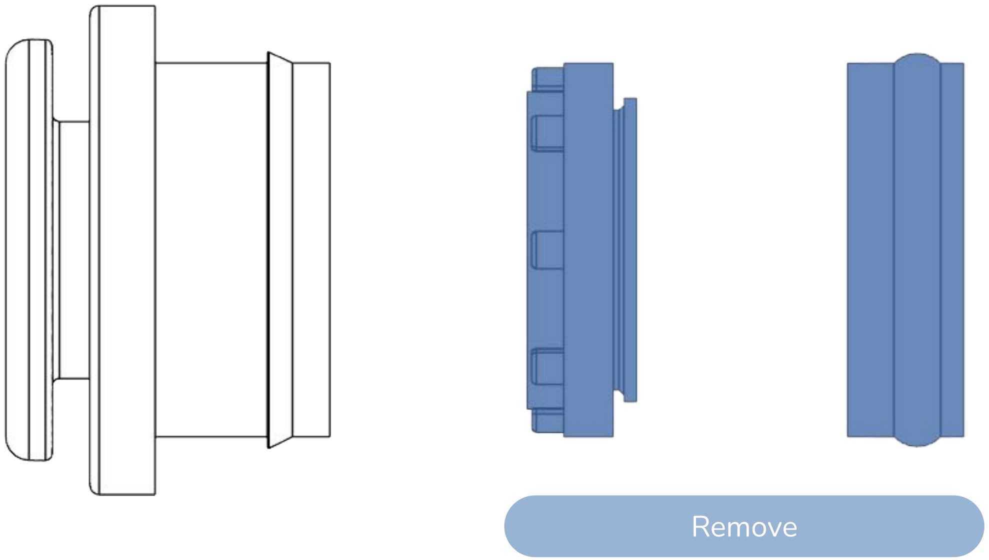
SHRINKAGE COMPENSATION
Parts are not compensated for filament shrinkage.
Calibrate your filament shrinkage compensation in the slicer before printing.

INFILL TYPE
Gyroid or Cubic

INFILL PERCENTAGE
Recommended: **40%**

WALL COUNT
Recommended: **4**
One Wall Top and Bottom surface

SOLID TOP/BOTTOM LAYERS
Recommended: **5**

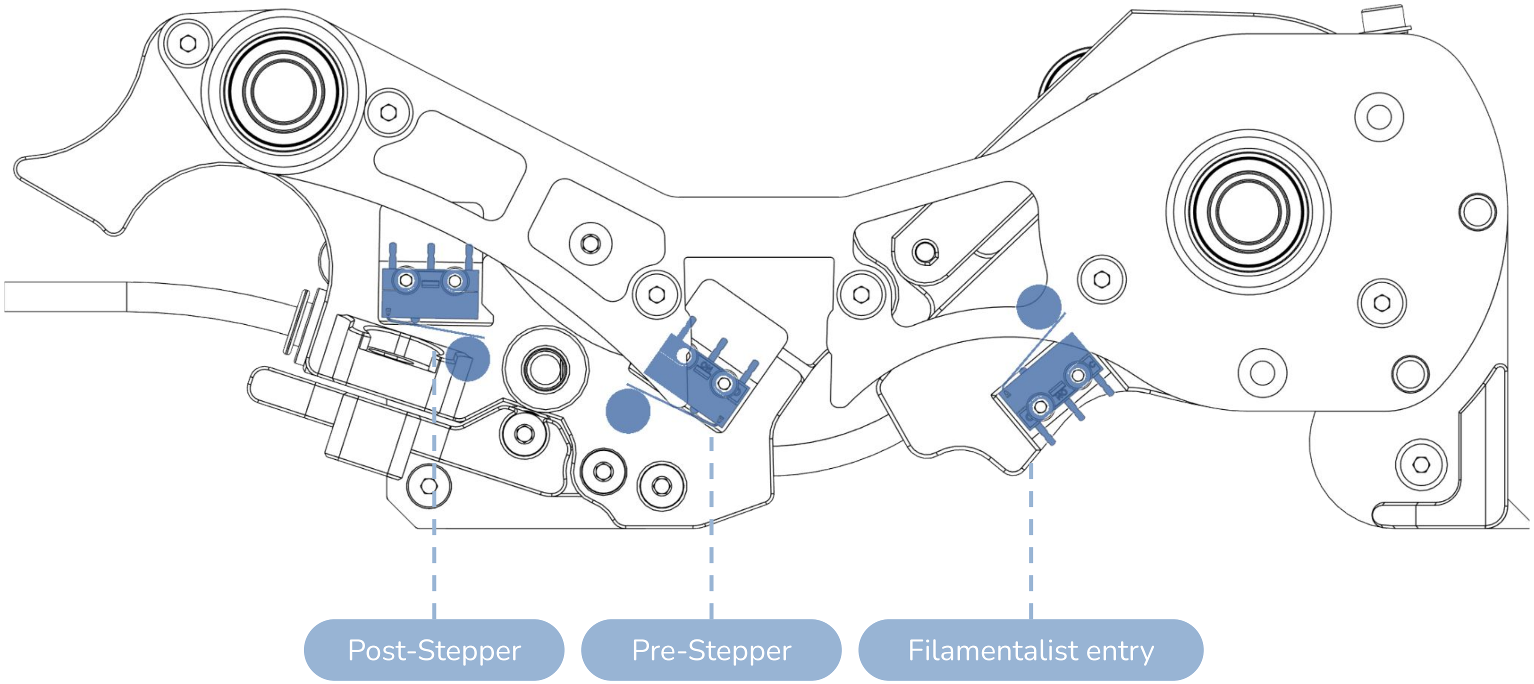


CAUTION:

Whenever we are using ECAS in this manual, the seal retainer and the seal should be removed

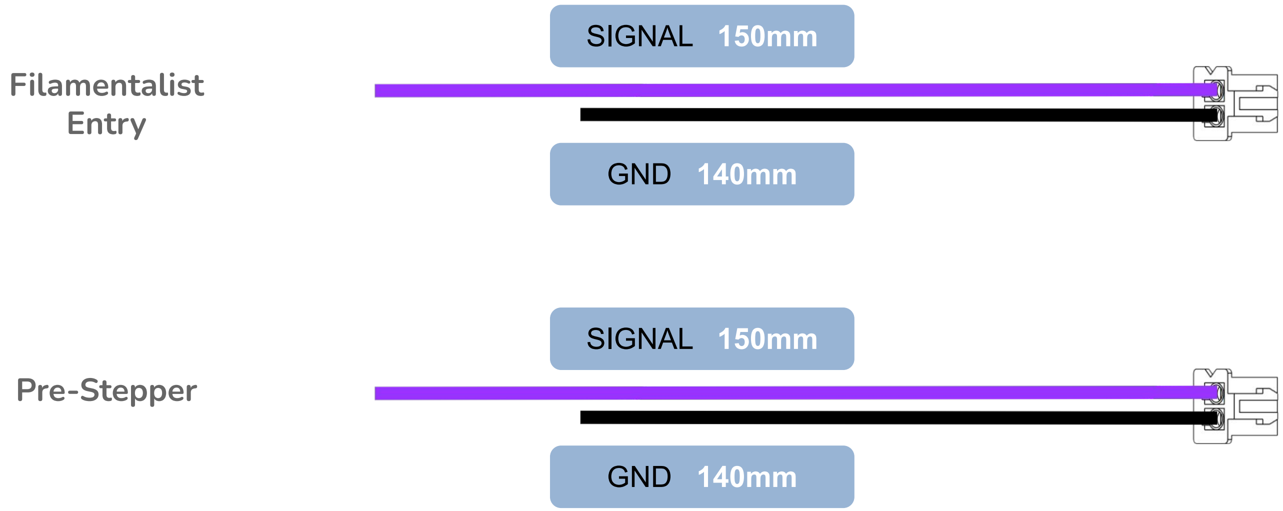
CHAPTER 1: WIRING PREPARATION

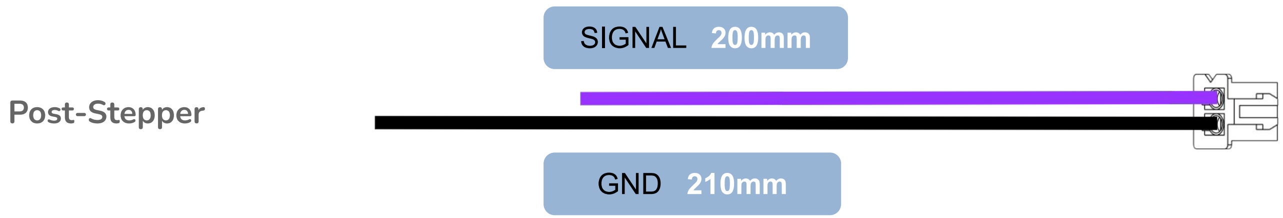
Start off by preparing the wiring for the filamentalist switch sensors. Their assembled position and naming is shown below.

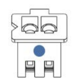


Step 1: Measure wires and trim to size

Step 2: If not pre-terminated, crimp JST XH terminal. Insert to JST XH plug





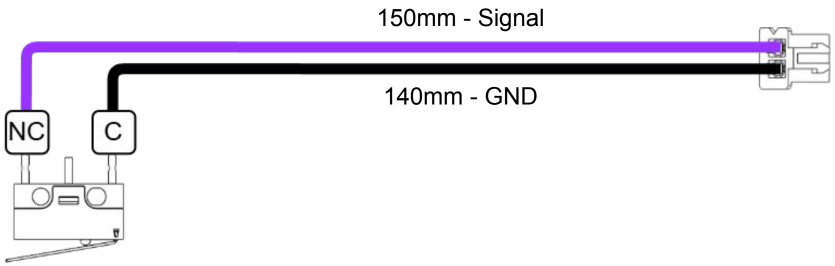


Tip : Mark the JST housing with a permanent maker to easily identify it later on

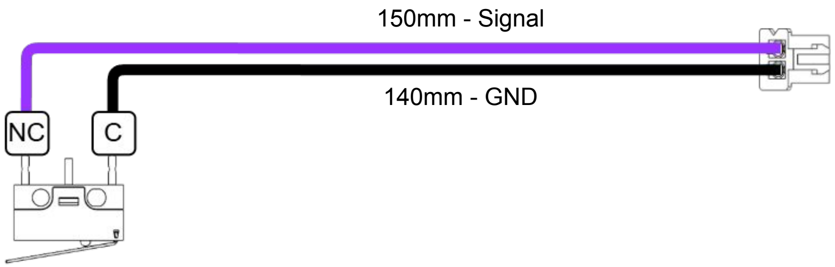
Depending on the source of your kit, the **JST XH terminals may already be installed on the wires.**
Wire orientation does not affect their functionality - ie C and NC can be swapped around.
Measurements are from the **end of the JST plug to the end of the wire.**

Step 3: Strip the insulation from the wire free end, pre-tin wires and switch legs and solder to the D2F-L switch outer two terminals as shown below.

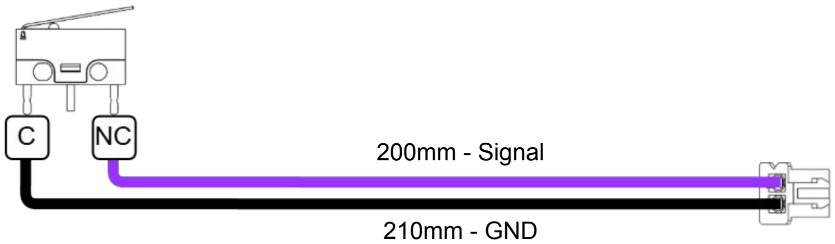
Filamentalist Entry

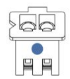


Pre-Stepper



Post-Stepper





Tip : Mark the JST housing with a permanent maker to easily identify it later on

CAUTION:
Pay attention to the switch and lever orientation!

Step 1: Measure wires and trim to size

Step 2: If not pre-terminated, crimp JST XH terminal. Insert to JST XH plug

Step 3: Strip the insulation from the wire free end, pre-tin wires and solder to the sensors as shown below.

Depending on the source of your kit, the **JST XH terminals may already be installed on the wires.**
Measurements are from the **end of the JST plug to the end of the wire.**

Step 1: Measure wires and trim to size

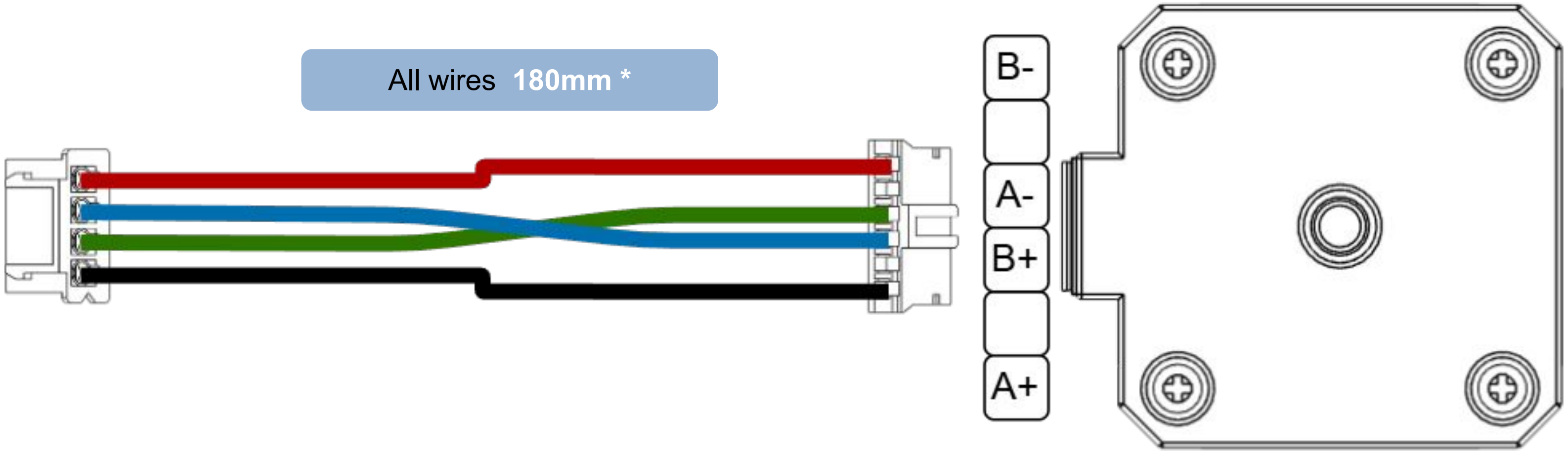
Step 2: If not pre-terminated, crimp JST XH terminal. Insert to JST XH plug

Step 3: Strip the insulation from the wire free end, pre-tin wires and solder to the sensors as shown below.

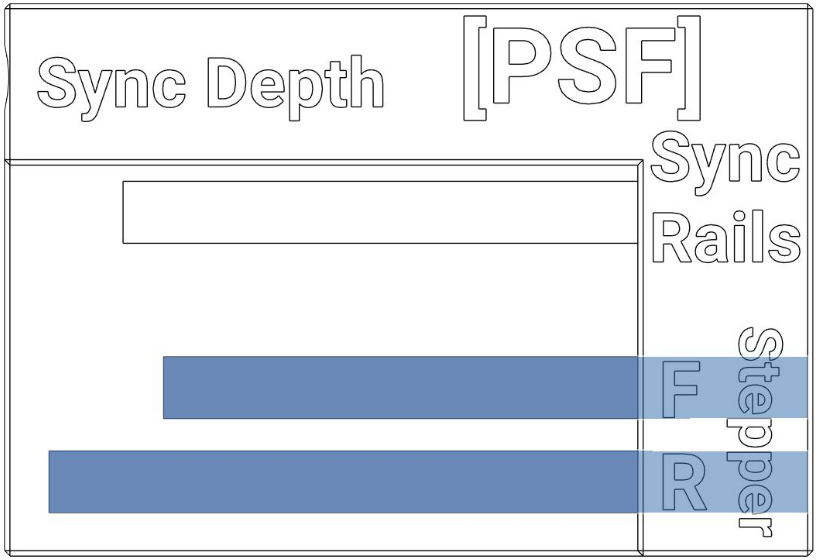
Depending on the source of your kit, the **JST XH terminals may already be installed on the wires.**
Depending on the source of your kit, the yellow wire may be “brown” or “orange”
Measurements are from the **end of the JST plug to the end of the wire.**

For the BOM steppers (OMC & Usongshine): The green and blue wires should be **swapped on stepper side** to match the AABB of the hatch board.

If using other steppers: refer to your stepper motor data sheet for its AABB pin output order.

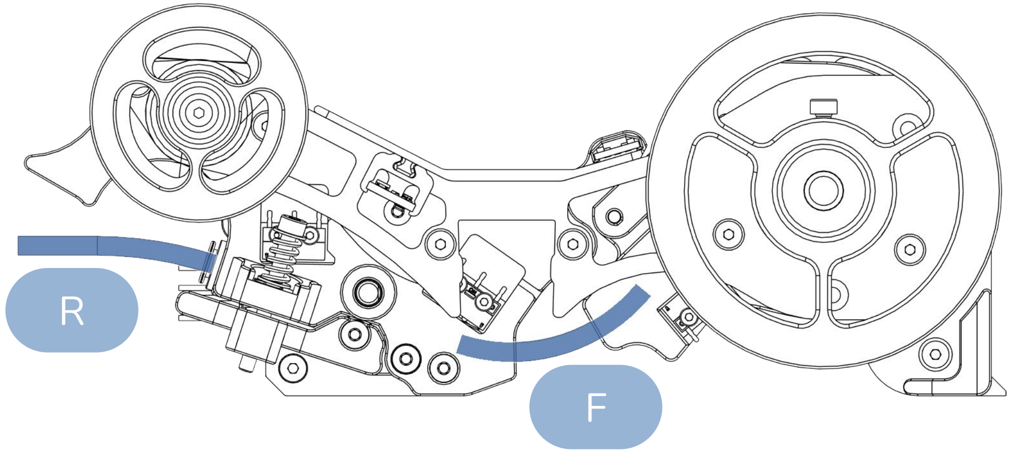


Depending on the source of your kit, the **JST XH terminals may already be installed on the wires.**
Depending on the source of your kit, the wires may be pre-crimped at 200mm. That is perfectly fine. The 20mm difference is not important.
Measurements are from the **end of the JST plug to the end of the wire.**



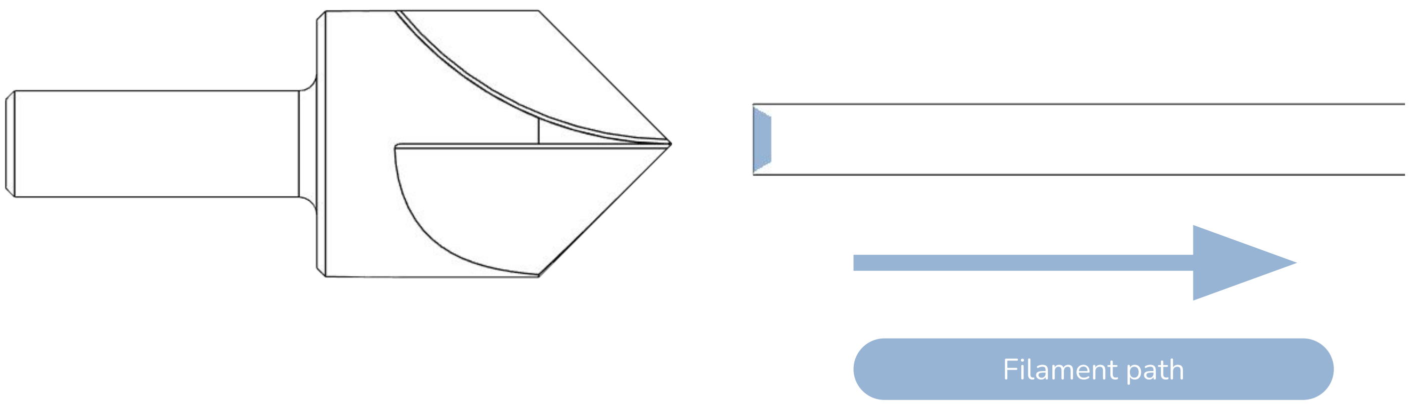
PTFE Cutter and Length Tool[PSF].stl

Cut here



Insert a **4mm OD, 2.5mm ID** piece of bowden tube in the cutting jig. **Cut flush with the body using a blade**. Alternatively mark with a pen and cut using a PTFE tube cutter.

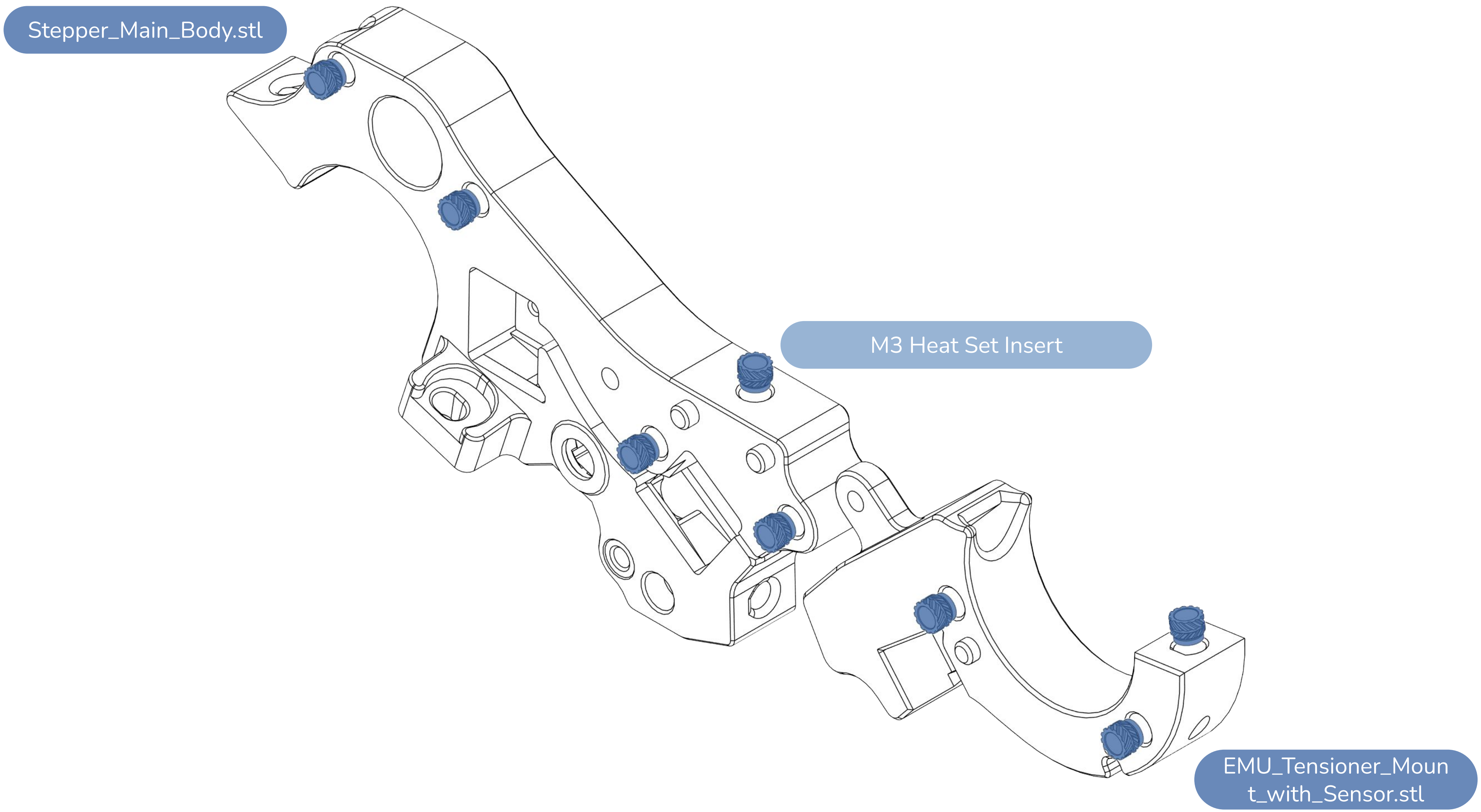
Cut one of each: front (F) and rear (R) tubes.



CAUTION:
To ensure smooth feeding of the filament, it is recommended that all **inlet sides of the PTFE tubes are slightly chamfered.** This can be done using a countersinking tool or a 4mm drill bit.

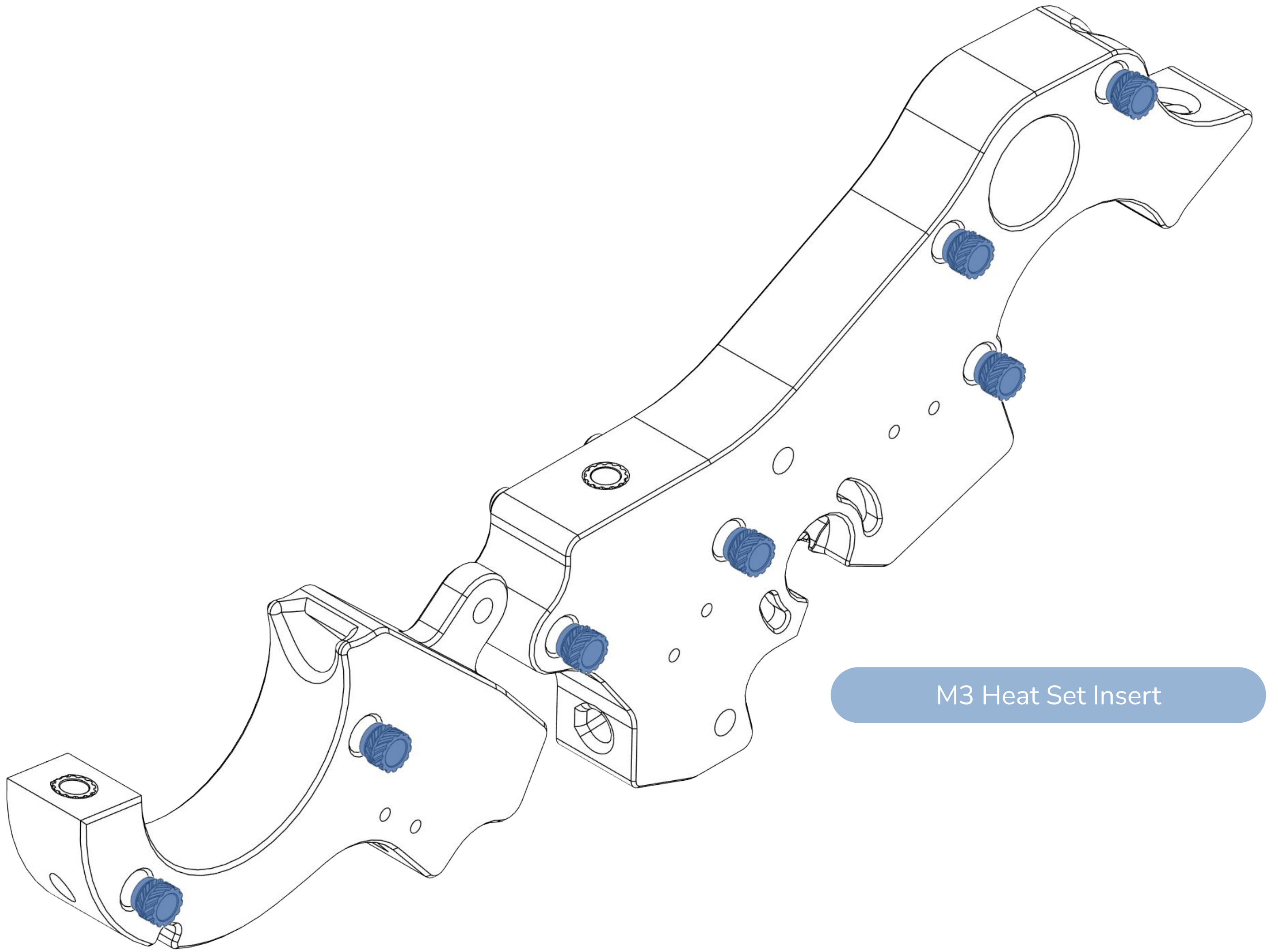
13

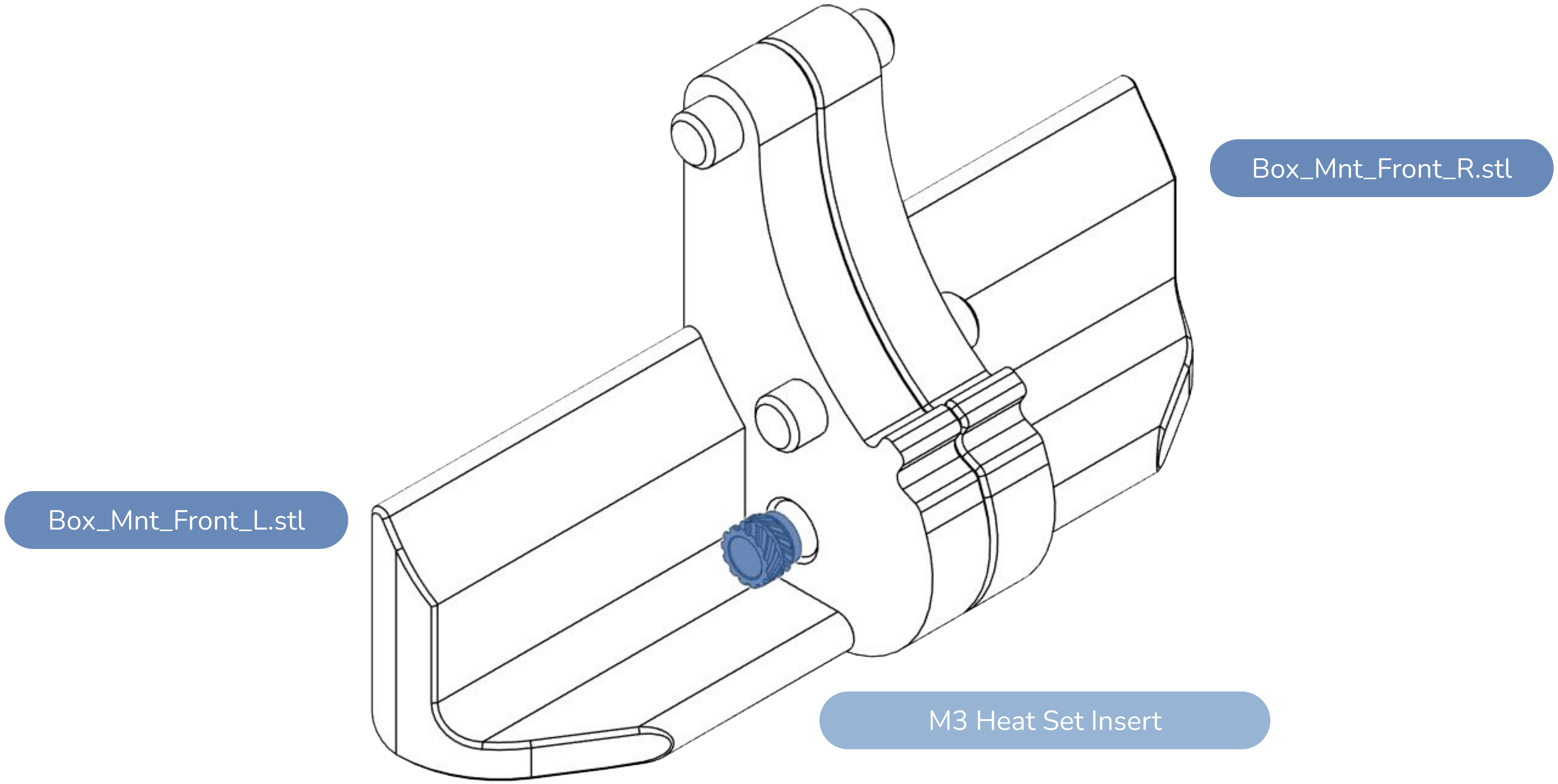
CHAPTER 2: FILAMENTALIST ASSEMBLY


Stepper_Main_Body.stl
M3 Heat Set Insert
EMU_Tensioner_Mount_with_Sensor.stl

15

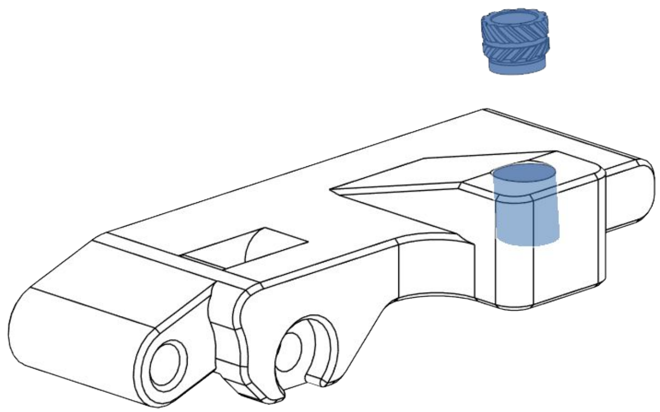
INSERT HEATSET INSERTS




Box_Mnt_Front_R.stl
Box_Mnt_Front_L.stl
M3 Heat Set Insert

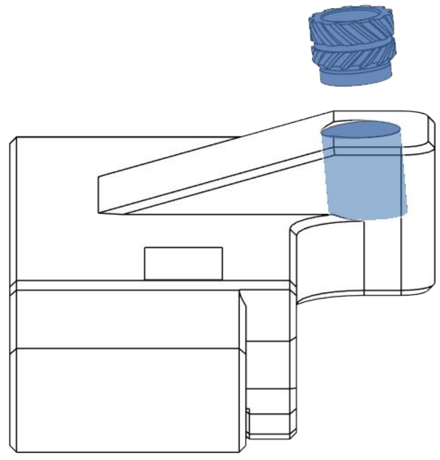
INSERT HEATSET INSERTS

M3 Heat Set Insert

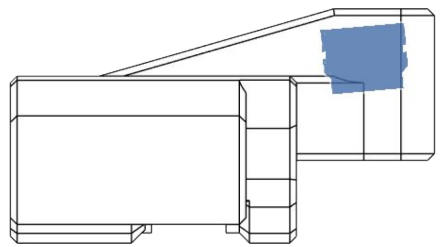


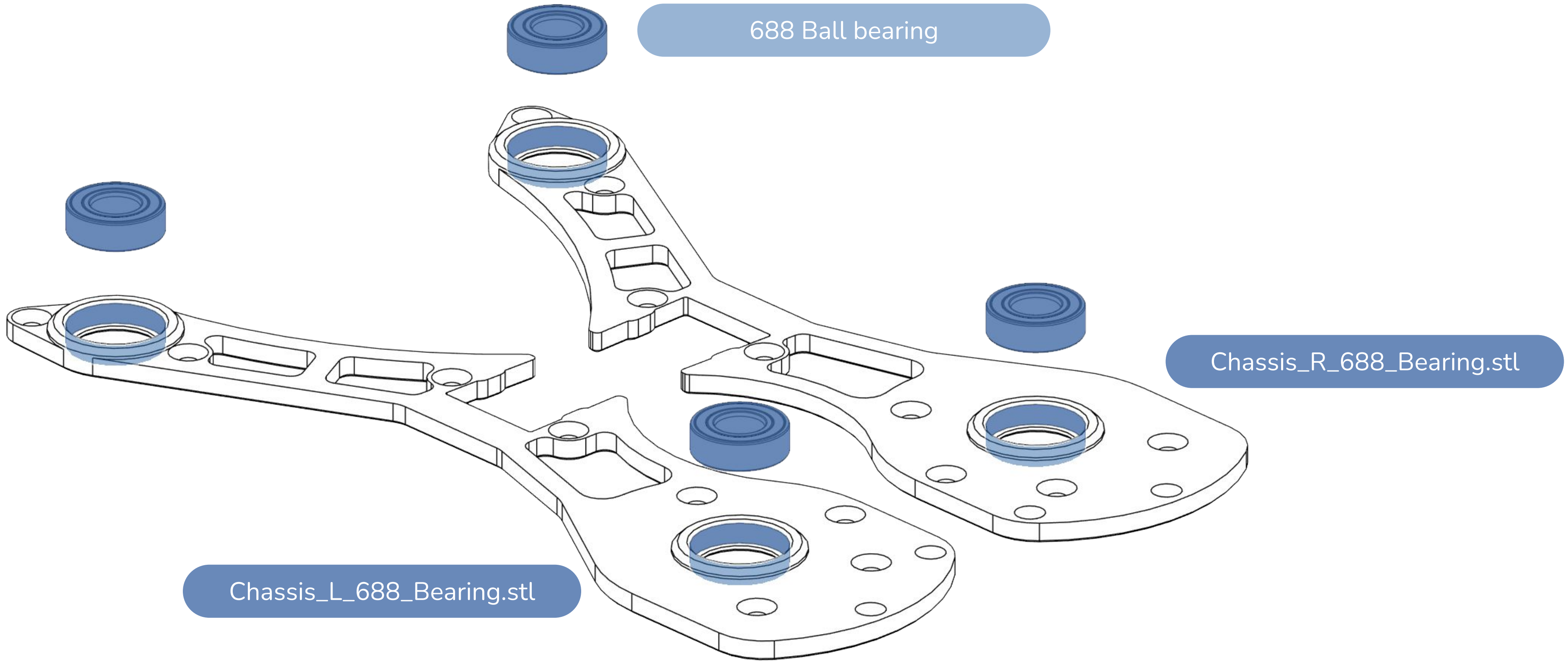
[a]_Stepper_Tension_Arm[MR693zz].stl

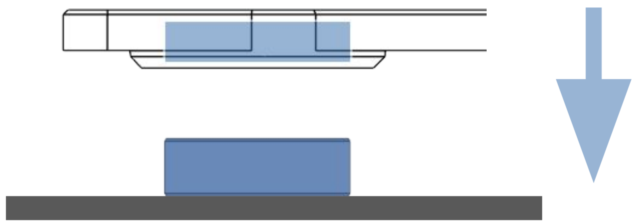
CAUTION:
The insert must at inserted at **an angle of 5°**



CAUTION:
The insert should be **recessed by ~1mm** to be fully seated



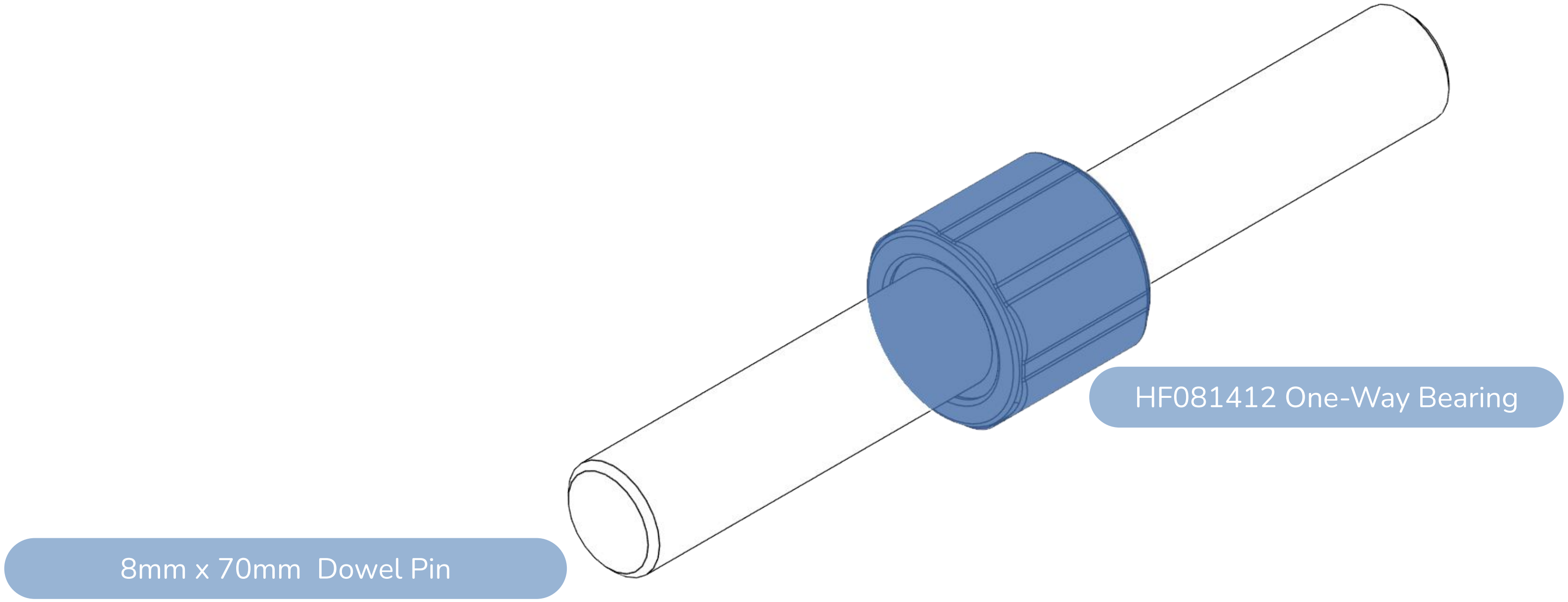




Lay the bearing on a flat surface and push down the frame so it’s easier to insert straight

19

TEST FIT ONE WAY BEARING



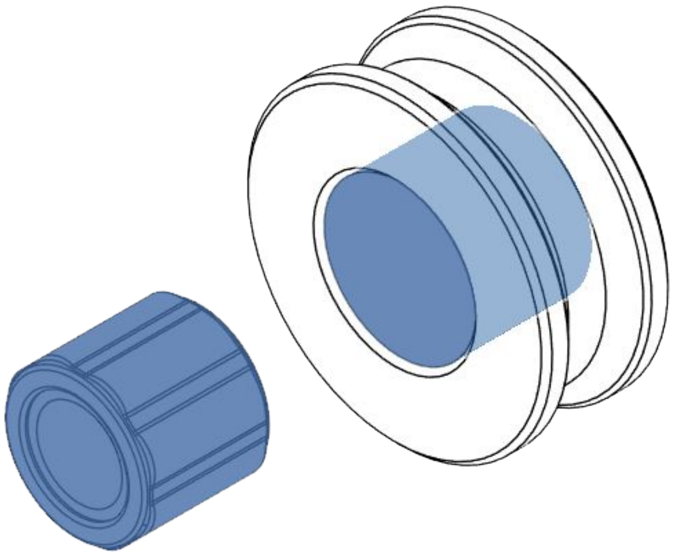
Test fit the one way bearing on the axle. Ensure it spins smoothly.

If it doesn't use a blowtorch to gently heat up the shaft, add the bearing while the shaft is hot and spin it around. This will adjust the inner locking mechanism tolerance and give a smoother rotation.

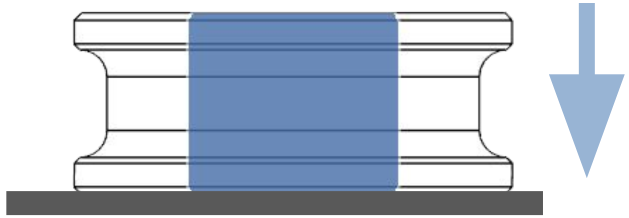
Once done, remove the one way bearing from the shaft and set aside.

INSERT ONE WAY BEARING INTO CDR

[a]_CDR.stl OR TPU_CDR/[a]_CDR.stl



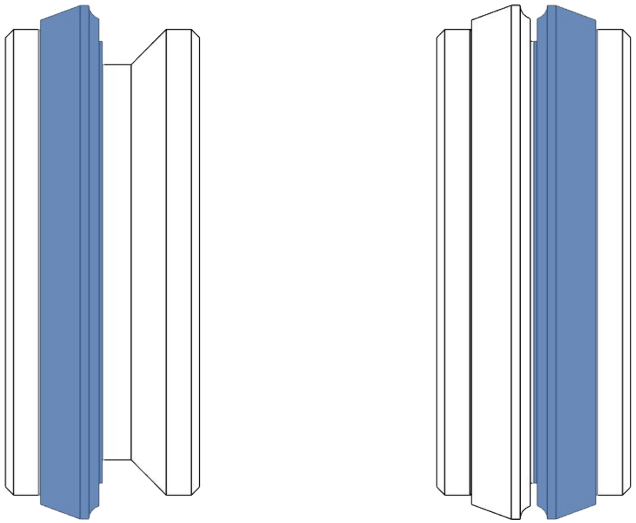
HF081412 One-Way Bearing



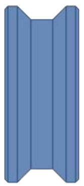
Lay the bearing on a flat surface and push the CDR down till flush with the surface.

ADD THE O-RINGS TO THE CDR

TPU Printed

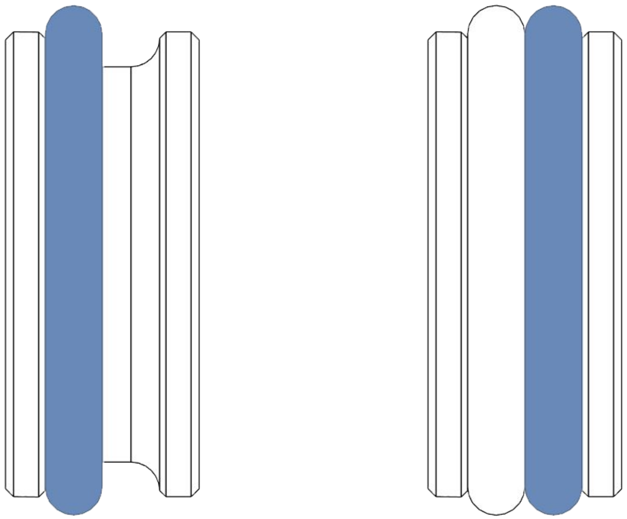


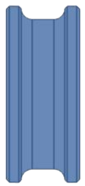
[TPU]_Half_Ring_Flex_1.65mm_x2.stl



TPU_CDR/[a]_CDR.stl

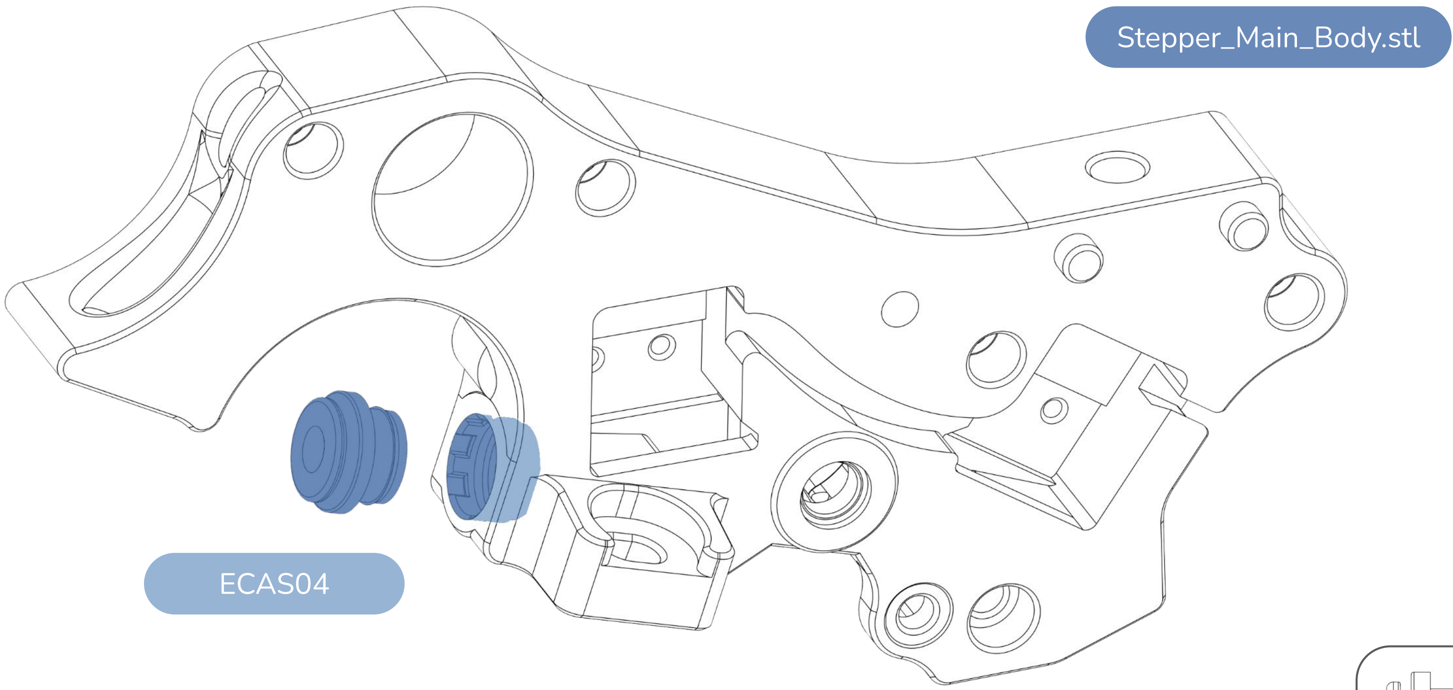
NBR O-Ring

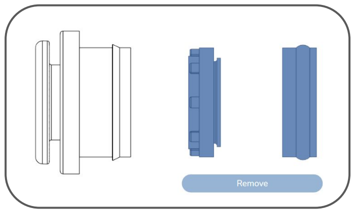




[a]_CDR.stl

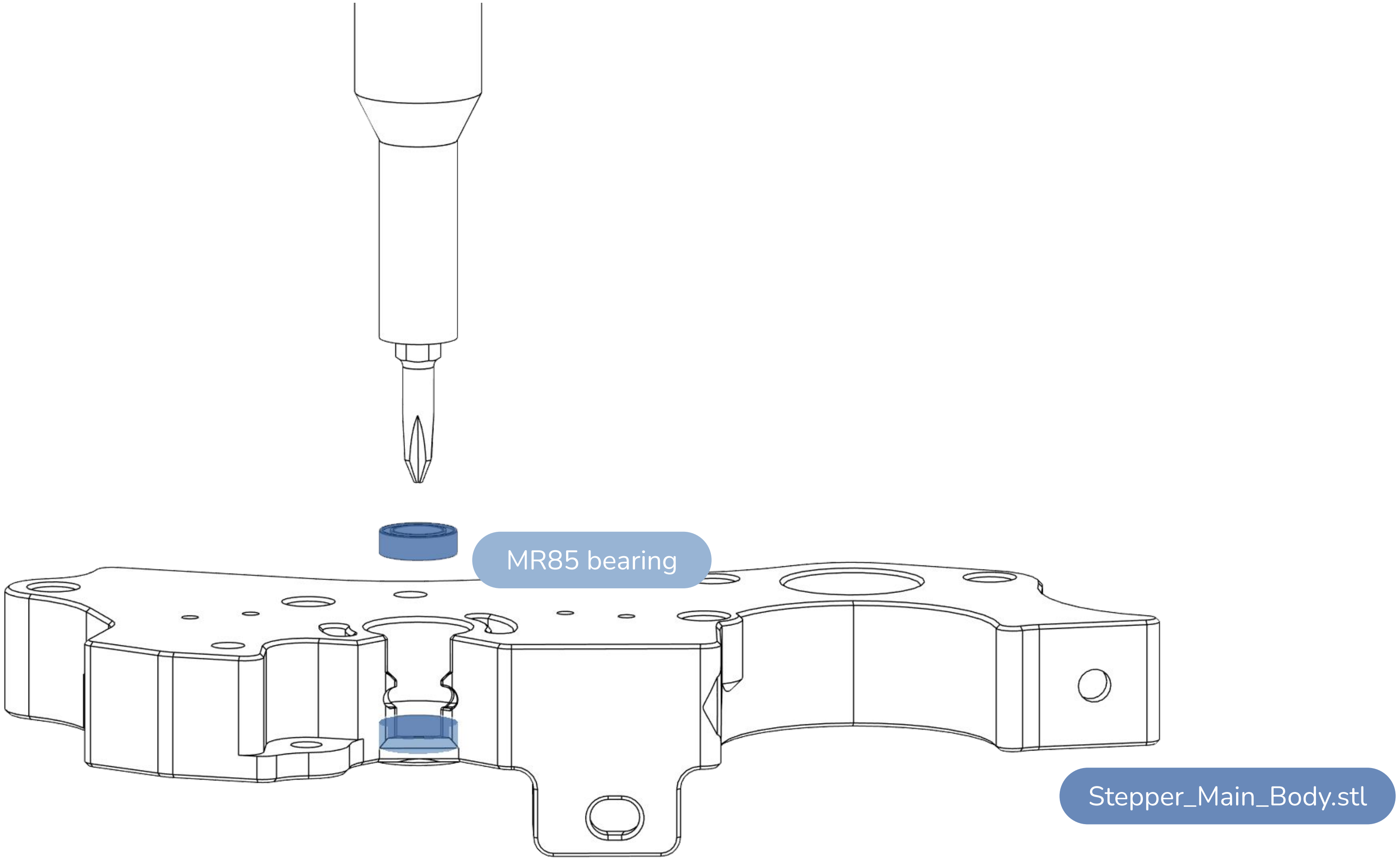
INSERT ECAS FITTINGS IN MAIN BODY





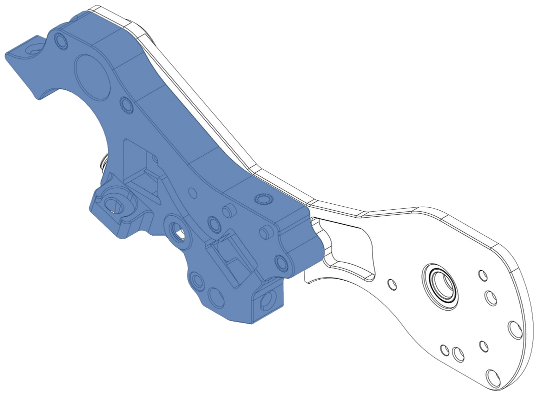
CAUTION:
Remember to remove and discard the retainer and the seal

INSERT MR85 BEARING INTO MAIN BODY

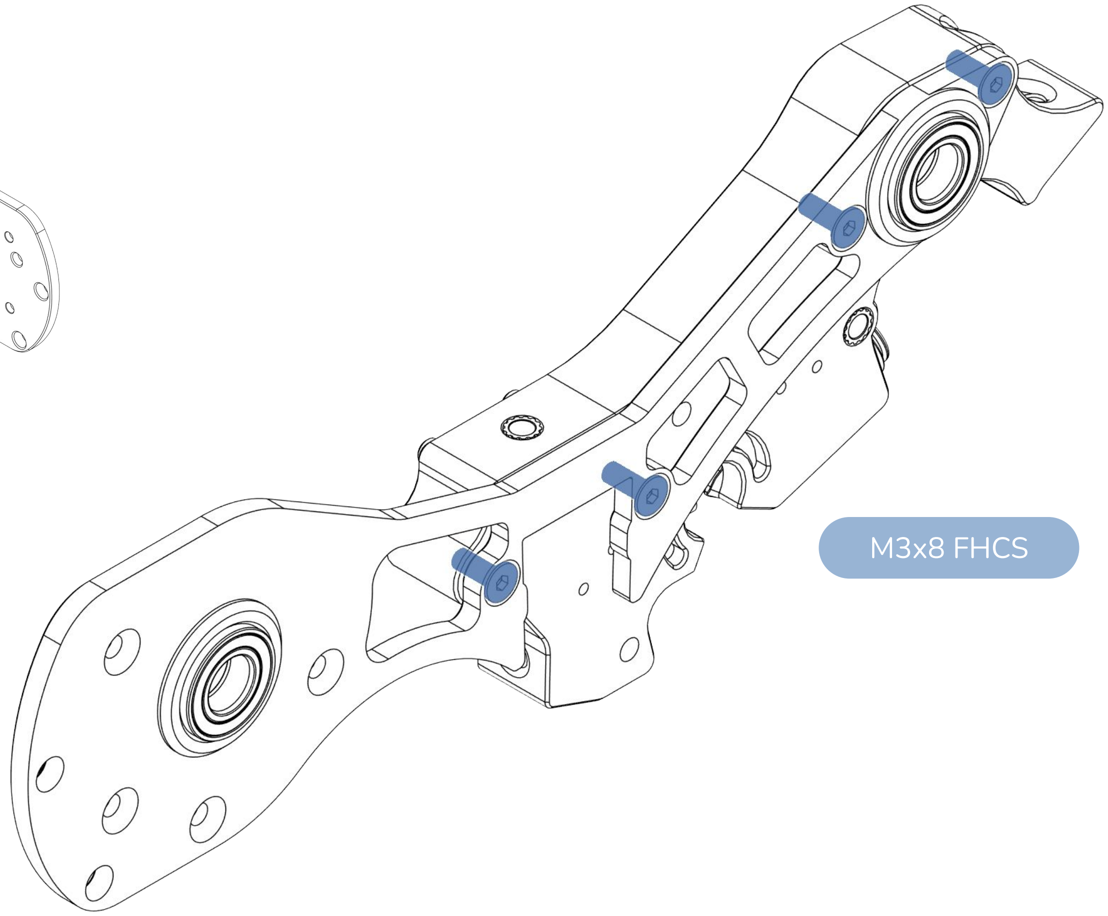


CAUTION:
Make sure the **bearing hole is clear of any loose bridges. Don't push too hard** as you may break the plastic part! Make sure the bearing is **fully seated.**

ASSEMBLE THE MAIN BODY TO THE CHASSIS

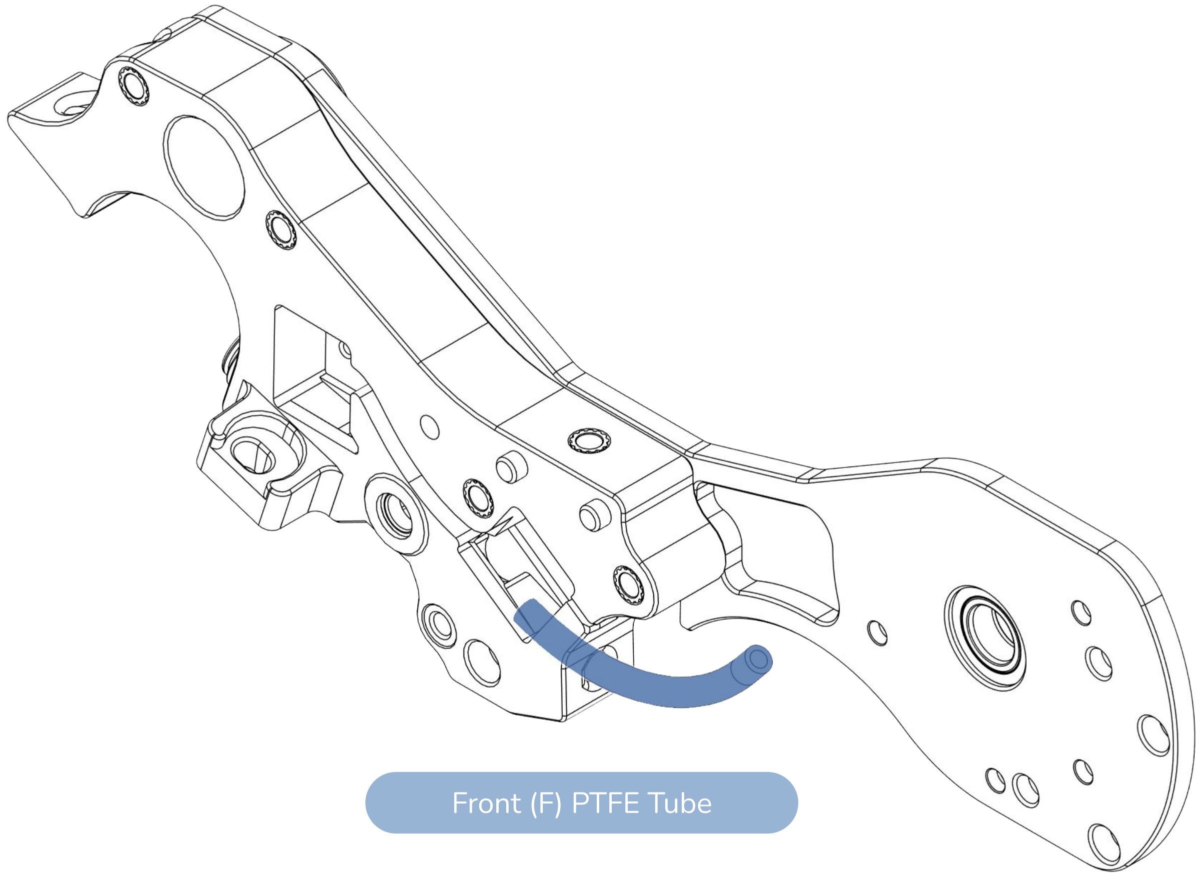


Stepper_Main_Body.stl

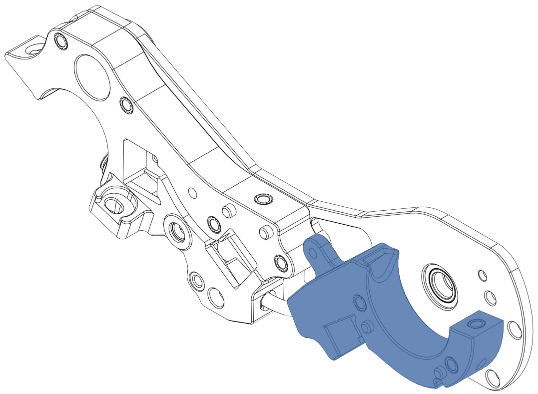


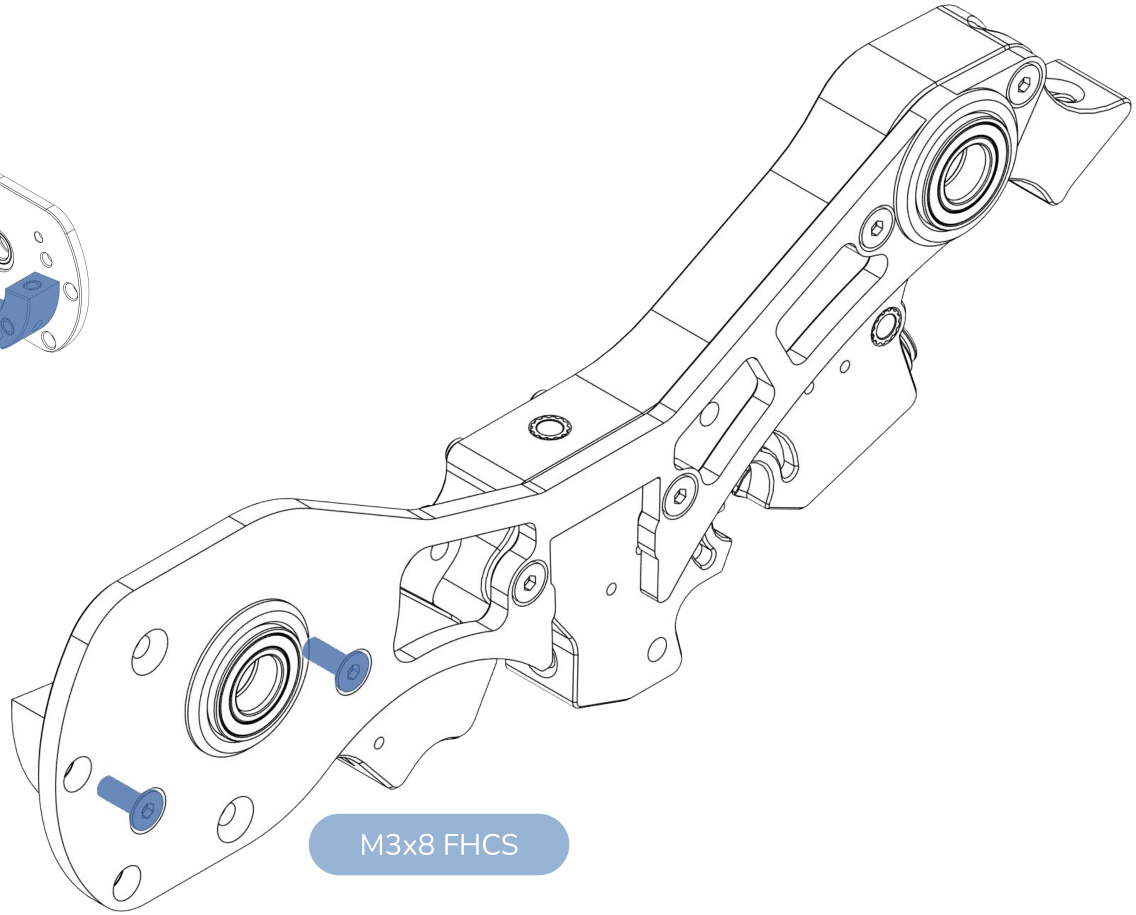
Chassis_L_688_Bearing.stl

INSERT THE FRONT PTFE TUBE

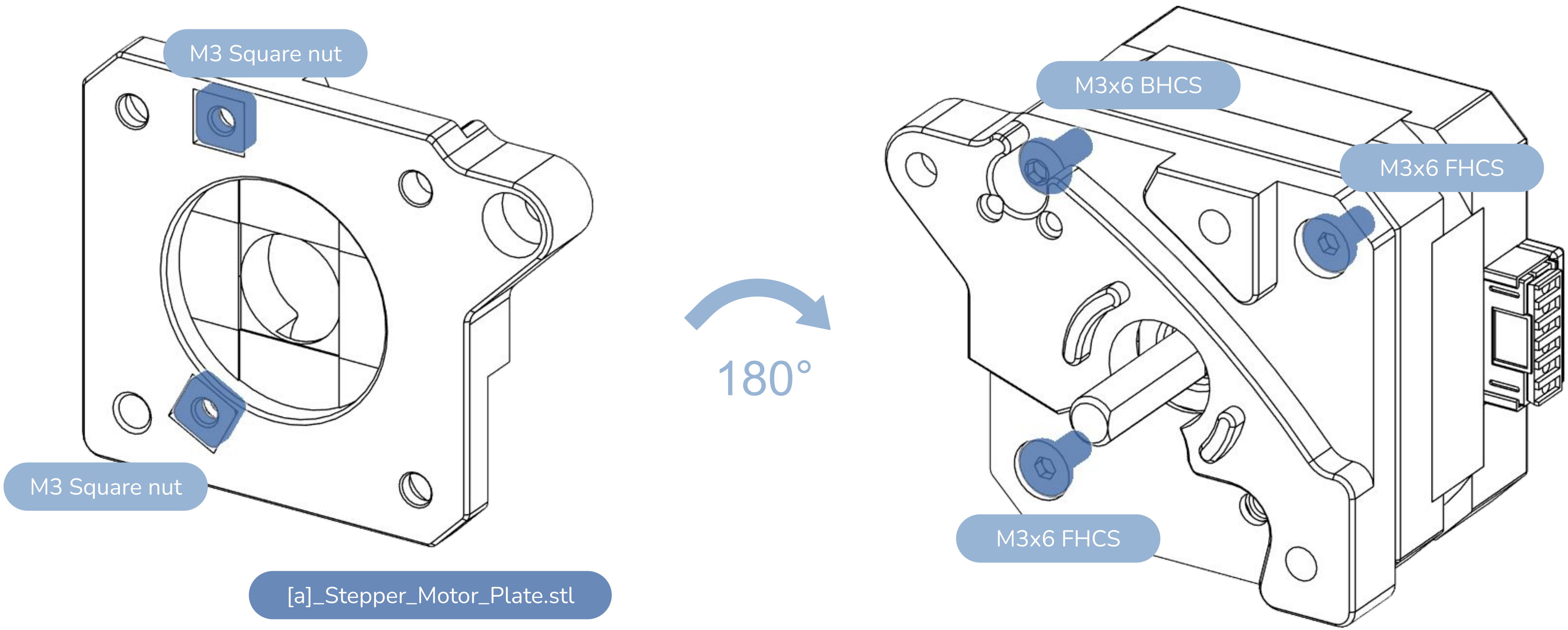


ASSEMBLE THE TENSIONER MOUNT TO THE CHASSIS

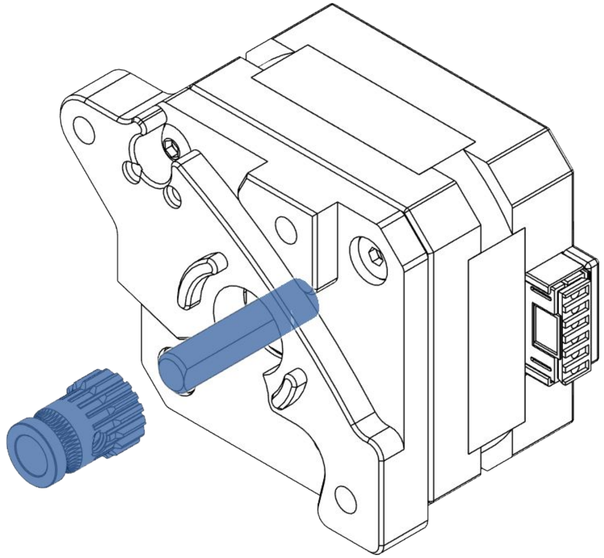




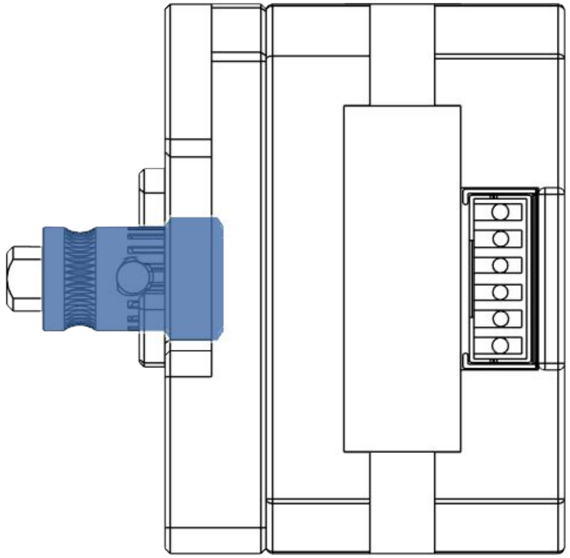
ASSEMBLE THE STEPPER MOTOR PLATE

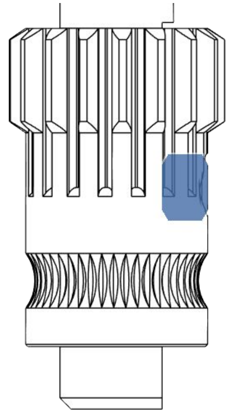


ASSEMBLE THE BMG GEAR & GRUB SCREW

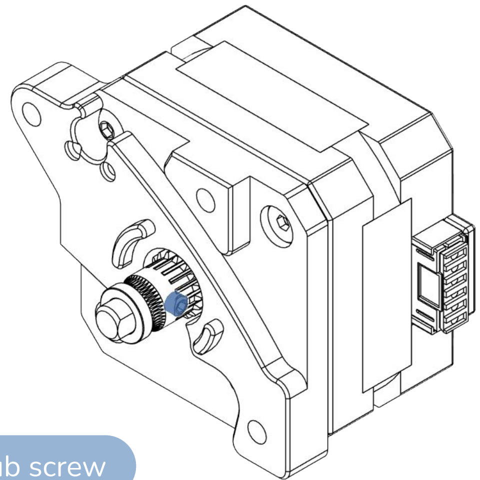


BMG Gear





Align the stepper shaft flat spot with the grub screw

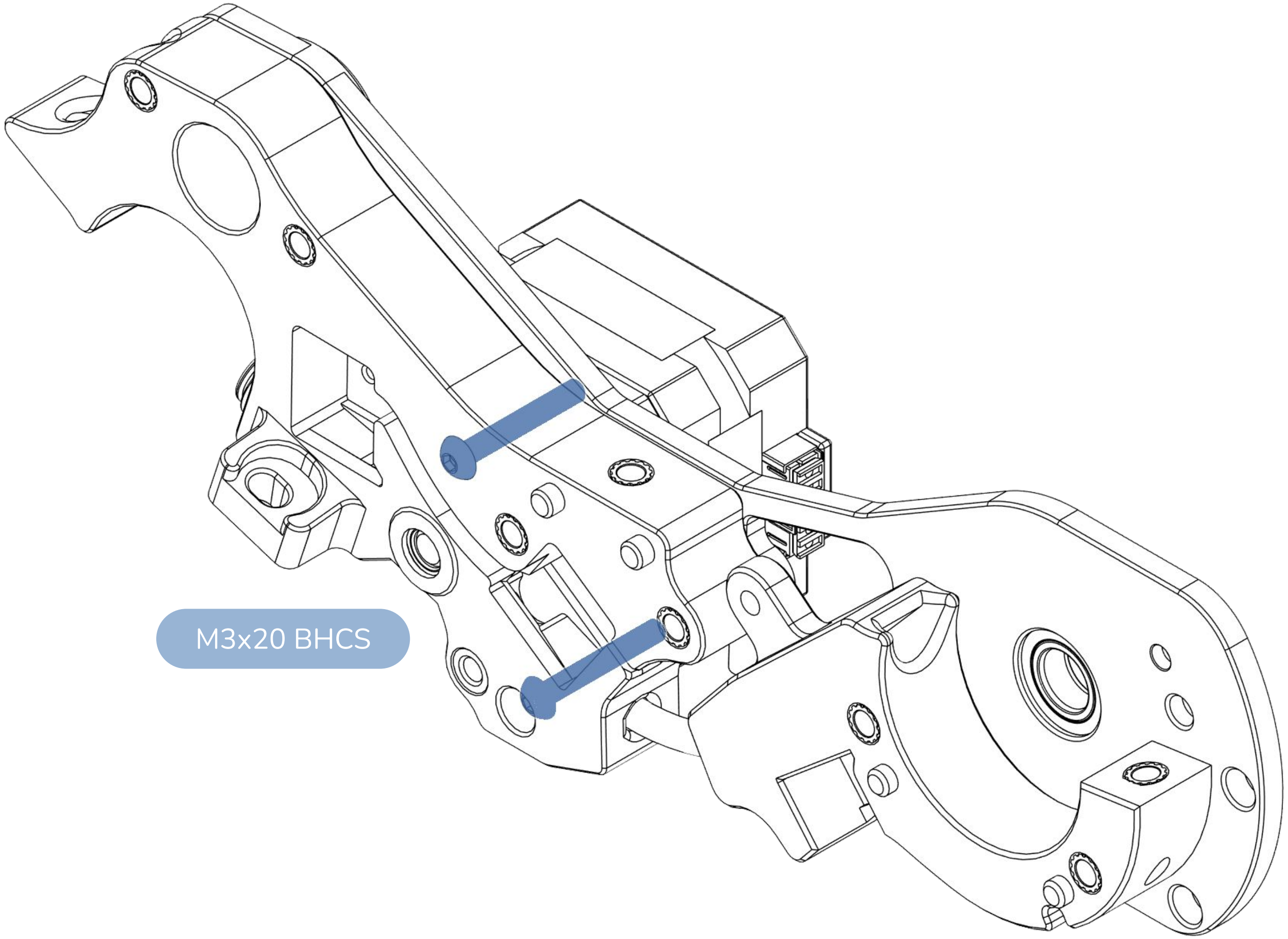


M2 Grub screw

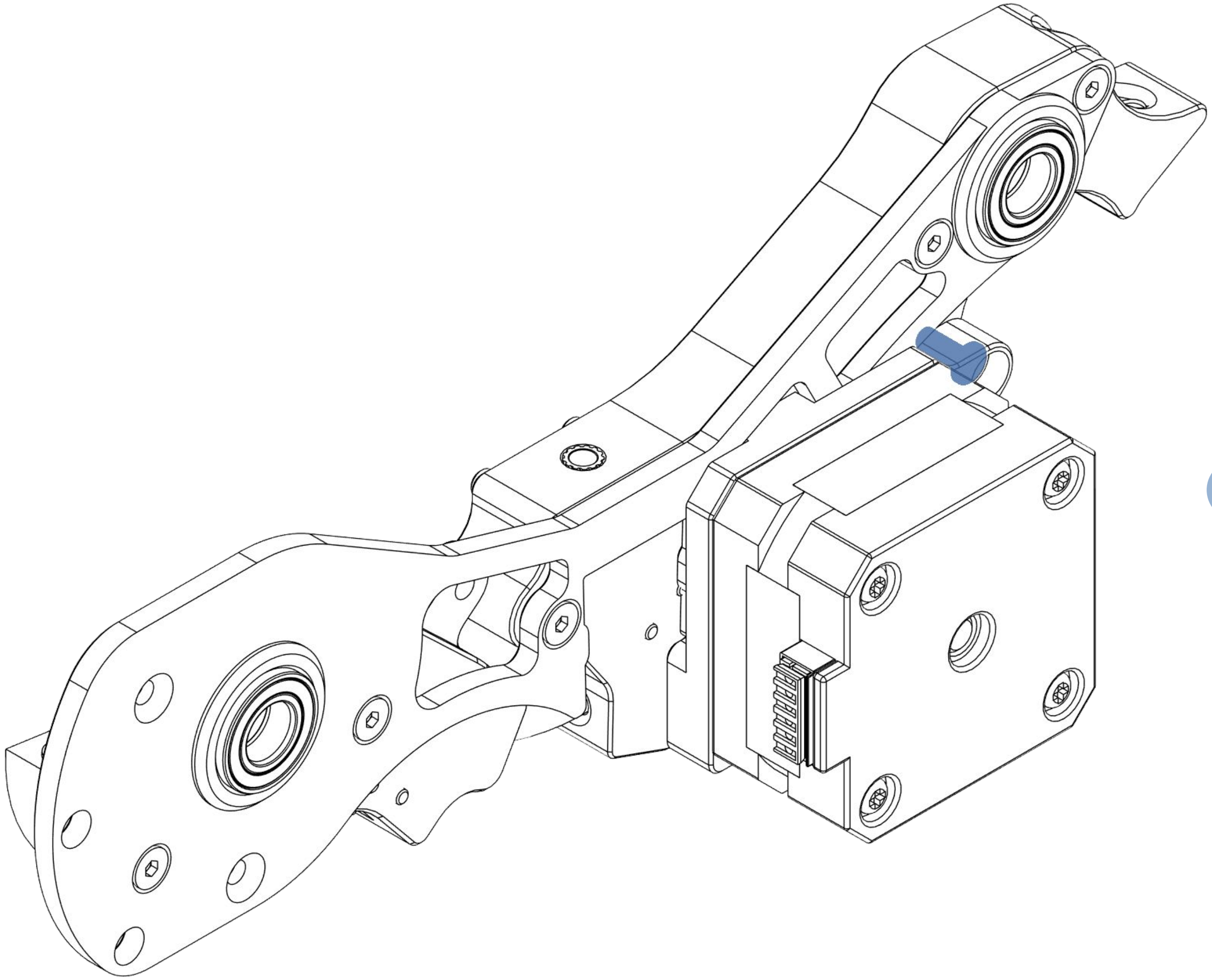
Tighten screw down, then unscrew by half a turn.

The BMG gear should be able to move up and down the shaft but not spin freely around the shaft.

ATTACH THE STEPPER TO THE MAIN BODY

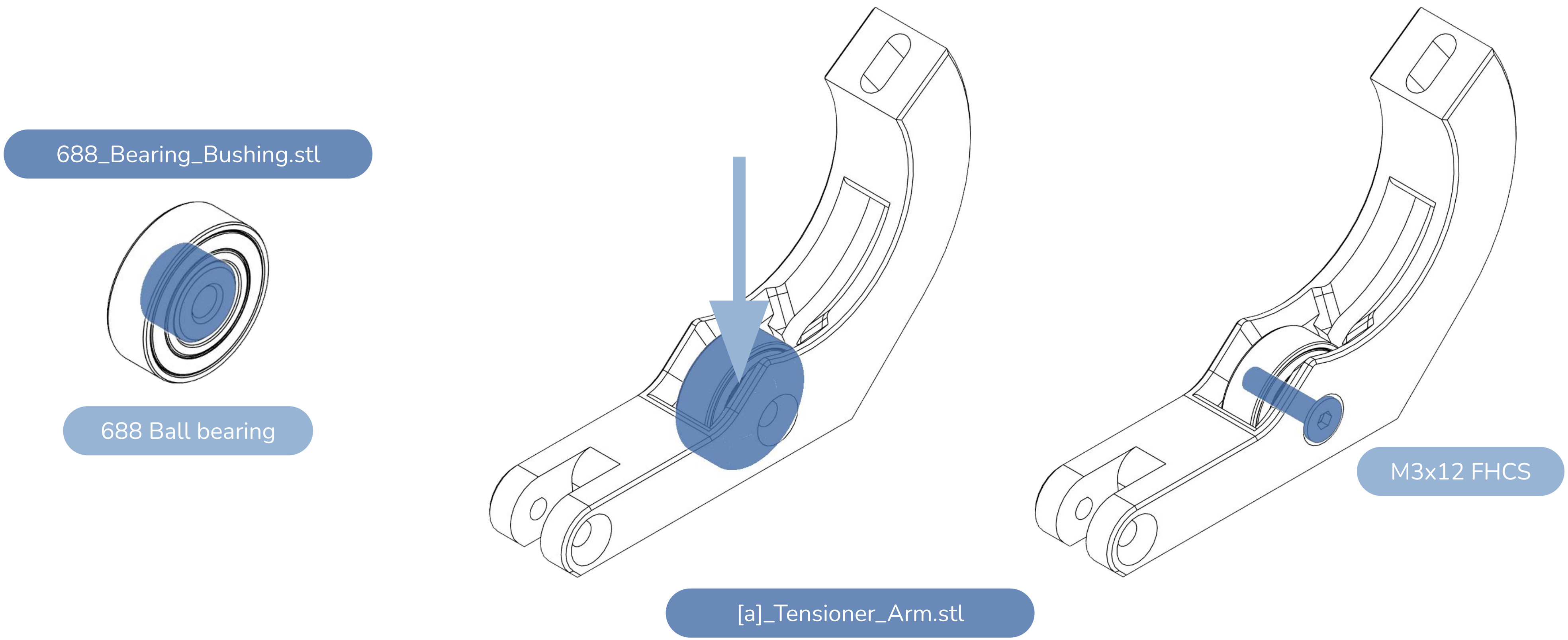


ATTACH THE STEPPER TO THE MAIN BODY



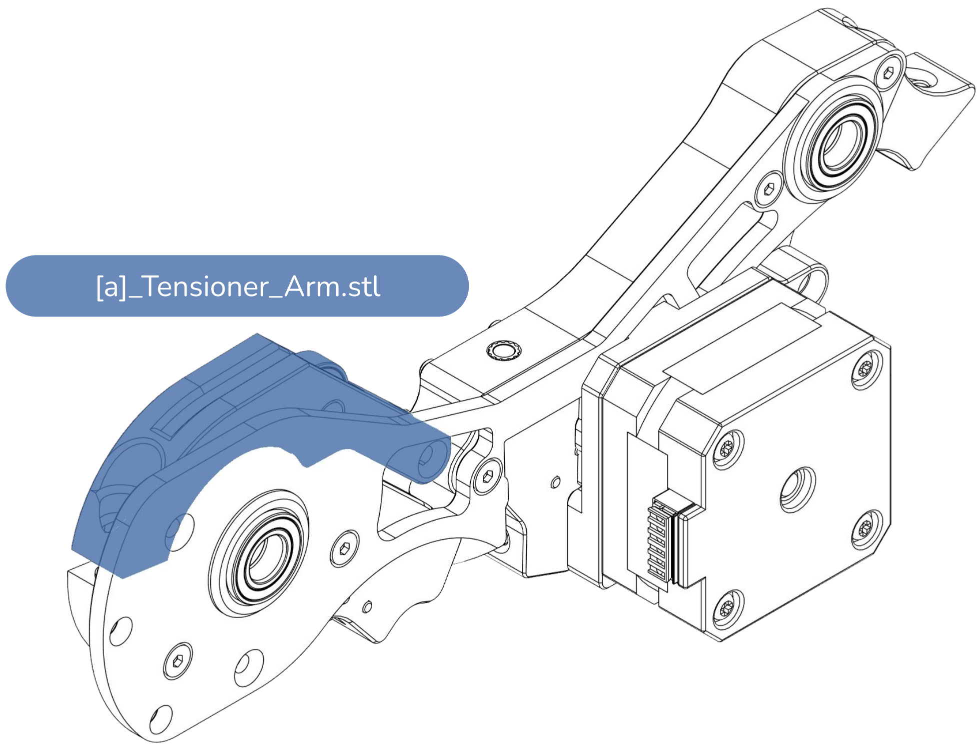
M3x8 FHCS

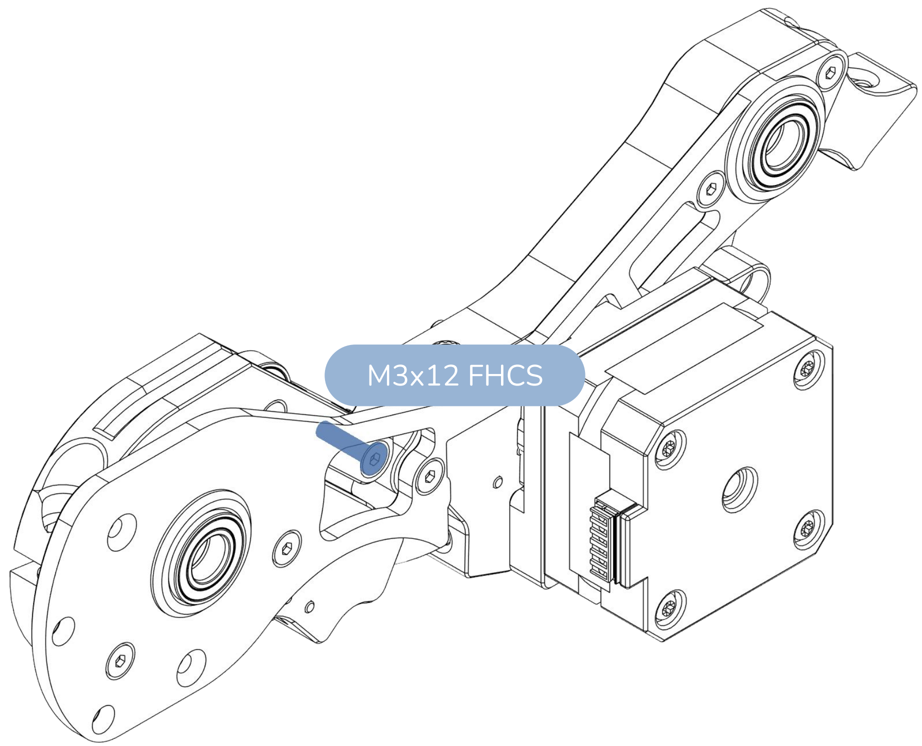
INSERT & SECURE THE TENSIONER ARM BEARING



CAUTION:
Be careful not to strip the plastic while securing the screw. The screw needs to be “finger tight” (snug).
Back the screw out by half a turn and **make sure the bearing spins freely**. The bearing may wobble side to side a little. That is perfectly fine.

ATTACH THE TENSIONER ARM TO THE ASSEMBLY



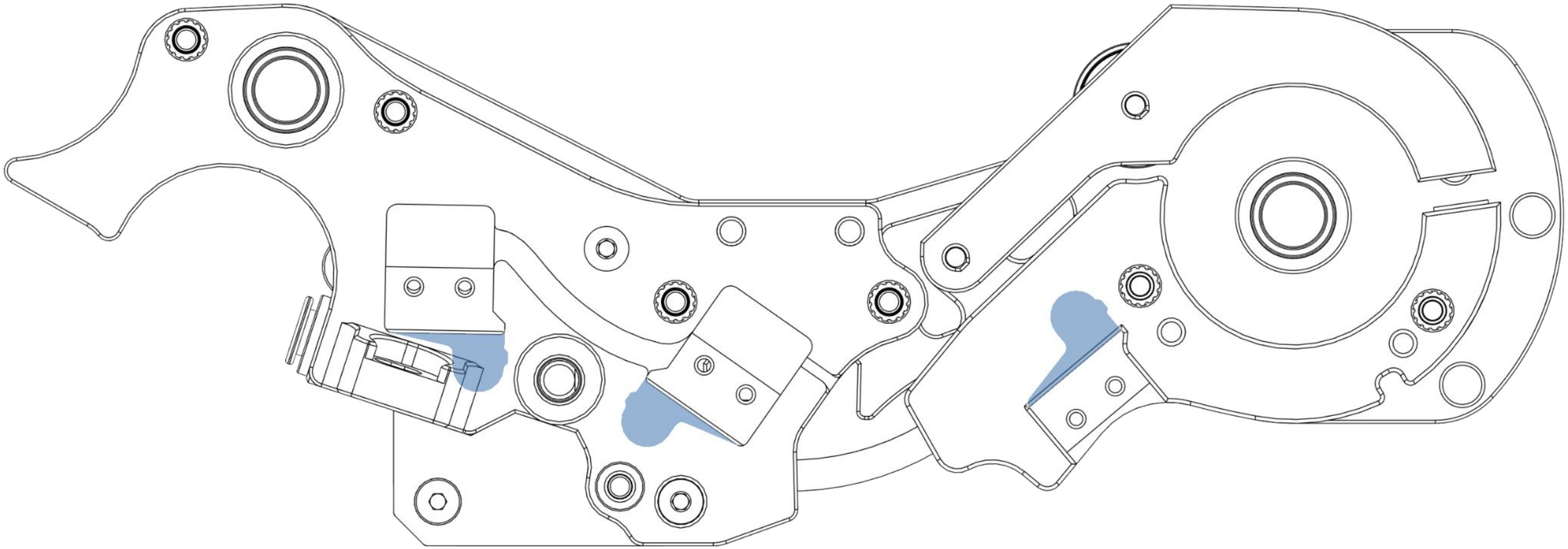


CAUTION:

Be careful not to strip the plastic while securing the screw. The screw needs to be “finger tight” (snug).

Back the screw out by half a turn and **make sure the arm flips freely.** It needs to feel loose. The arm may wobble side to side a little. That is perfectly fine.

TEST FIT THE SENSOR MAGNETS / DISKS



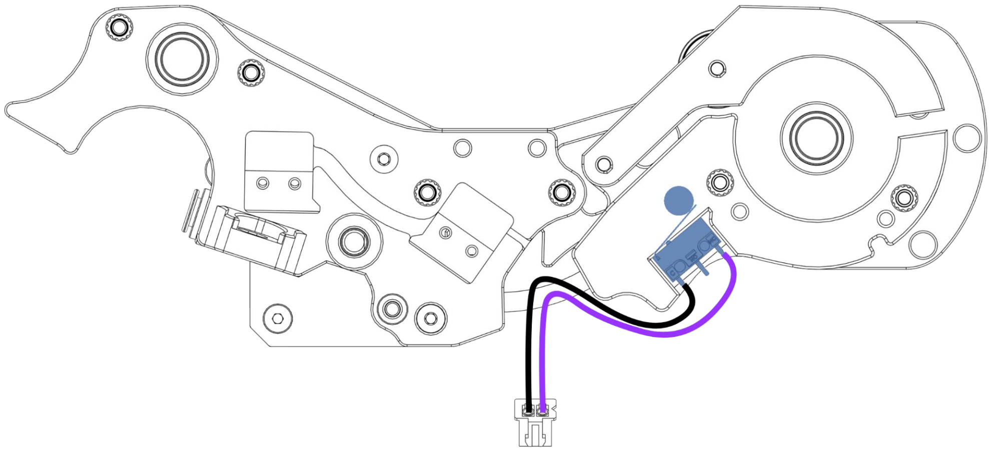
CAUTION:

The magnets must be loose enough to fall out on their own when you flip the assembly over. If they are tight this can cause sensor activation issues. This is caused by either incorrect shrinkage compensation or broken bridges when printing the part. You can alternatively self source and use a set of MR63 bearings in place of the magnets/disks if they are too tight.

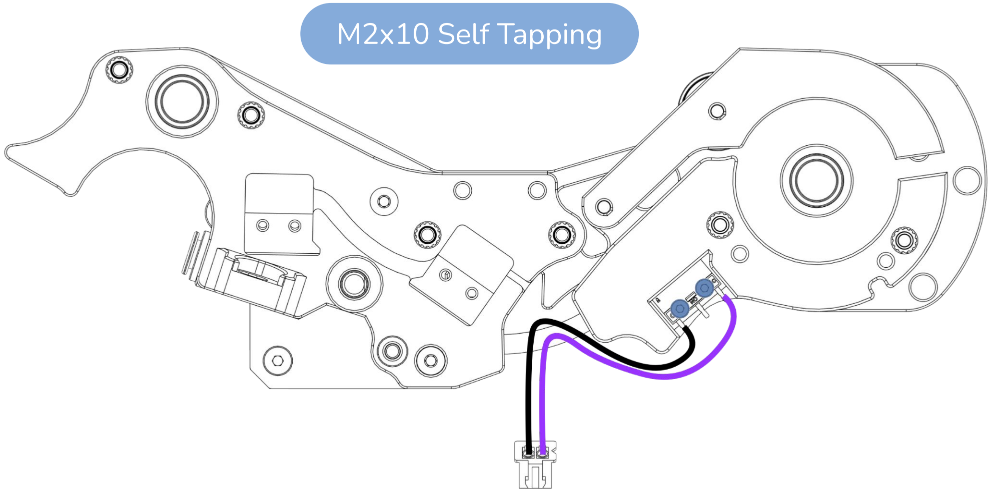
Depending on your kit source **the magnets may need to be demagnetised.** Use a blowtorch to gently heat them up till they lose their magnetism.

Install the entry sensor switch into the Filamentalist tensioner assembly.

Place the magnet/disk/bearing and align the switch as shown.



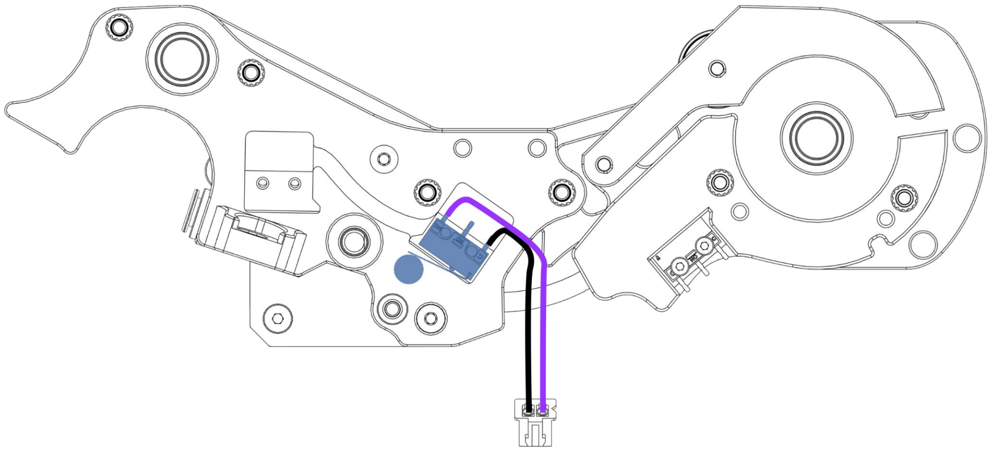
Install the two screws to secure the switch.
Careful not to overtighten!



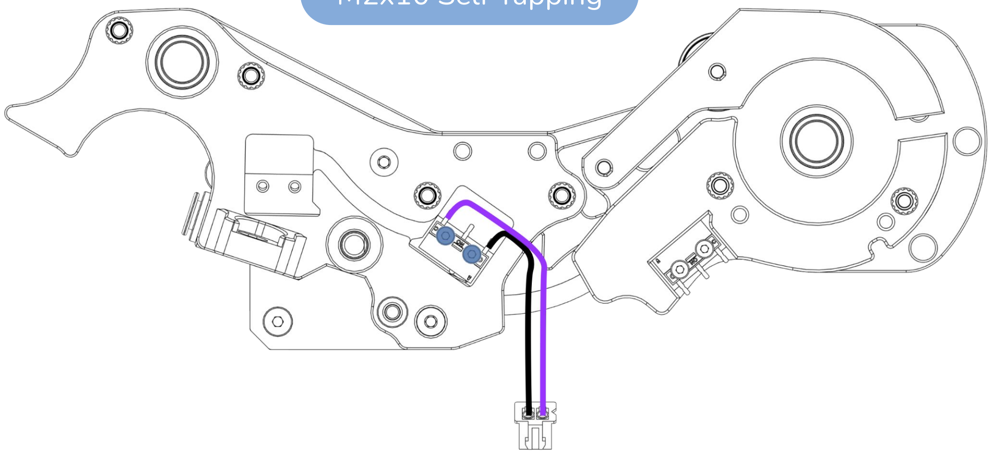
35

Install the pre-stepper sensor switch into the Filamentalist tensioner assembly.

Place the magnet/disk/bearing and align the switch as shown.

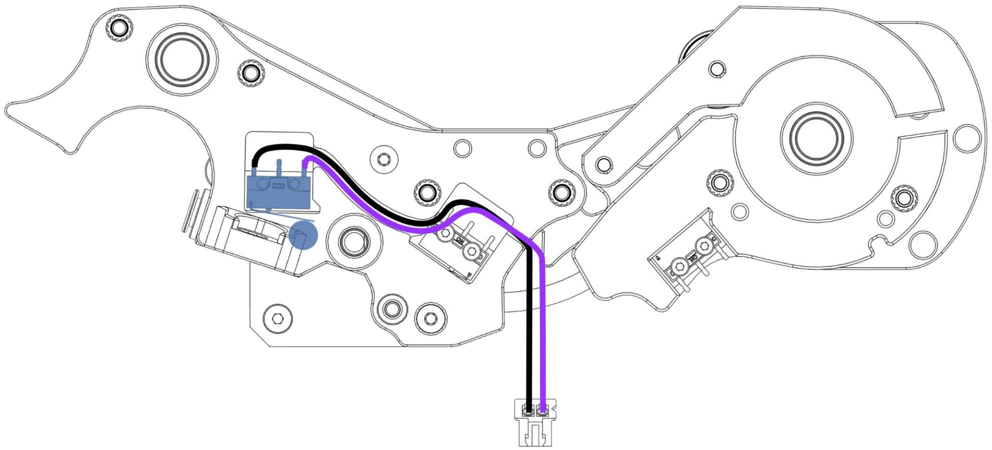


Install the two screws to secure the switch.
Careful not to overtighten!

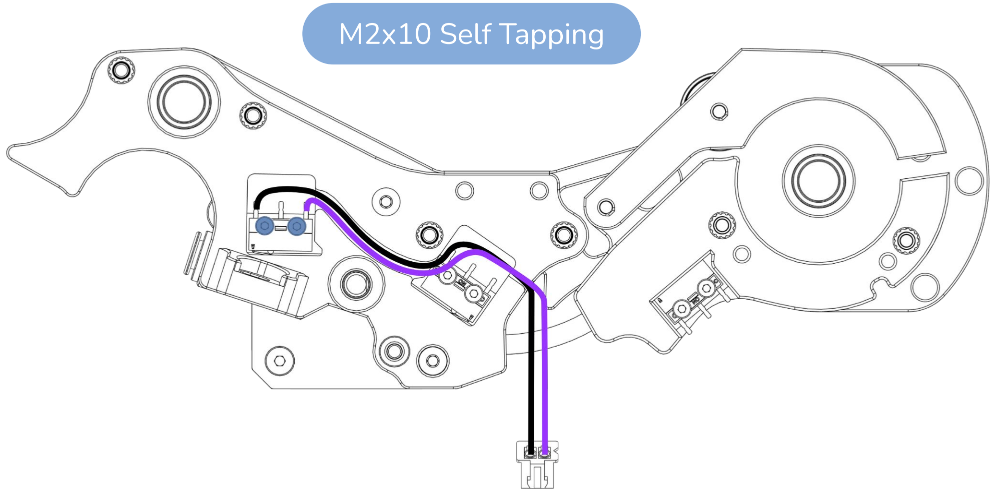


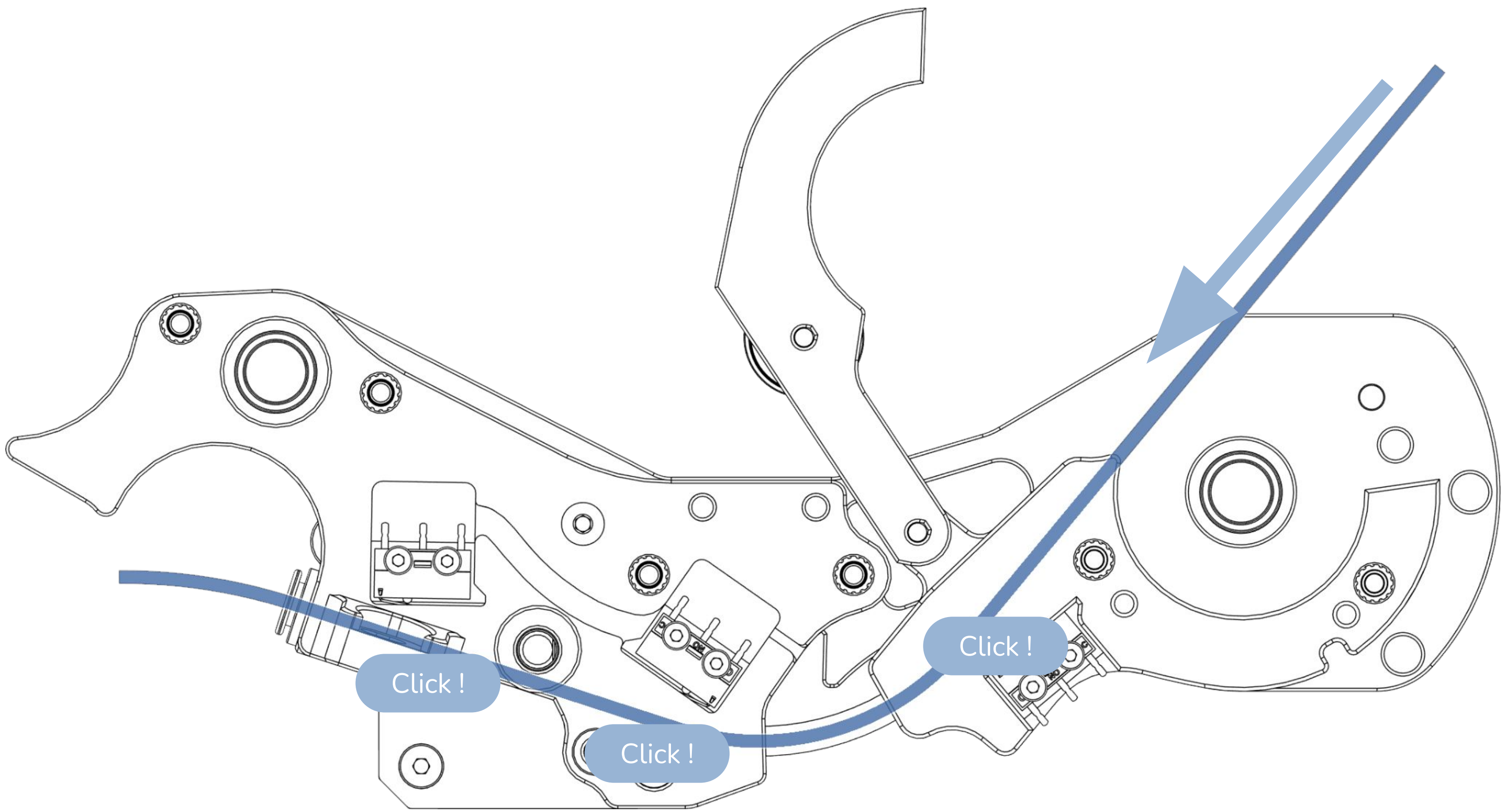
Install the post stepper sensor switch into the Filamentalist tensioner assembly.

Place the magnet/disk/bearing and align the switch as shown.



Install the two screws to secure the switch.
Careful not to overtighten!



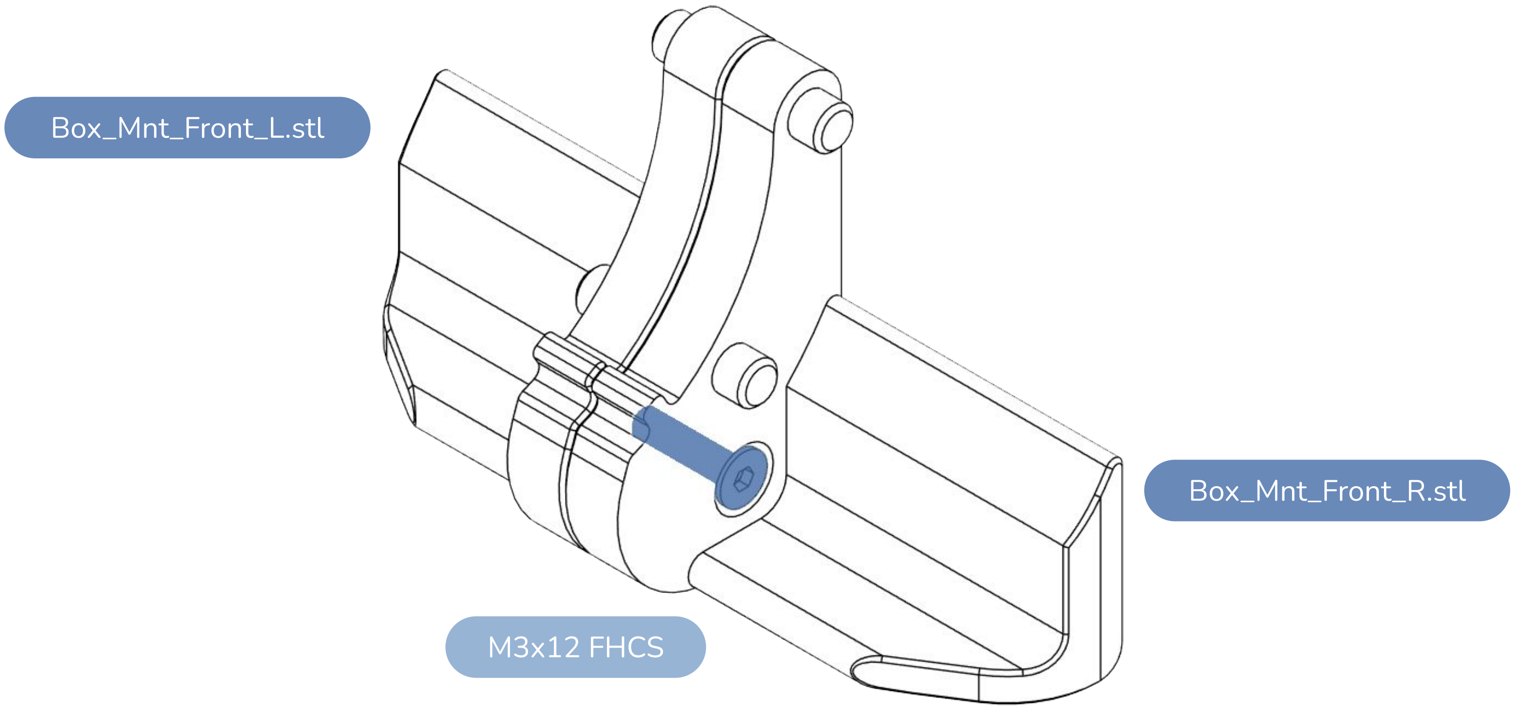


CAUTION:

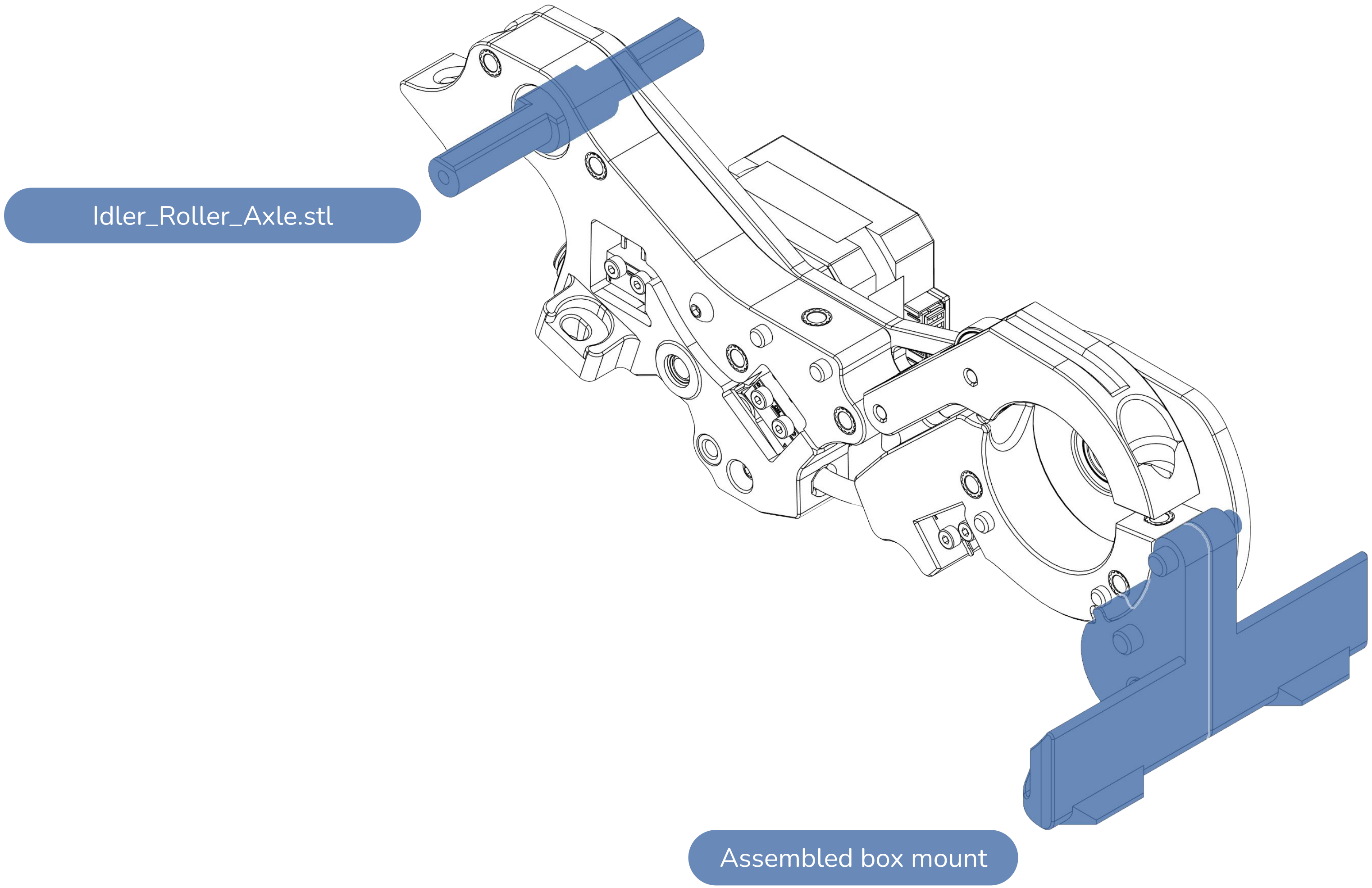
It is critical that the sensors trigger and release reliably. They are used to detect filament position and any incorrect activation will result in print issues.

Using some filament trigger the sensors individually a few times to ensure they trigger reliably. If not, ensure bridges are clean, loosen the M2 screws, adjust switch position slightly and tighten.

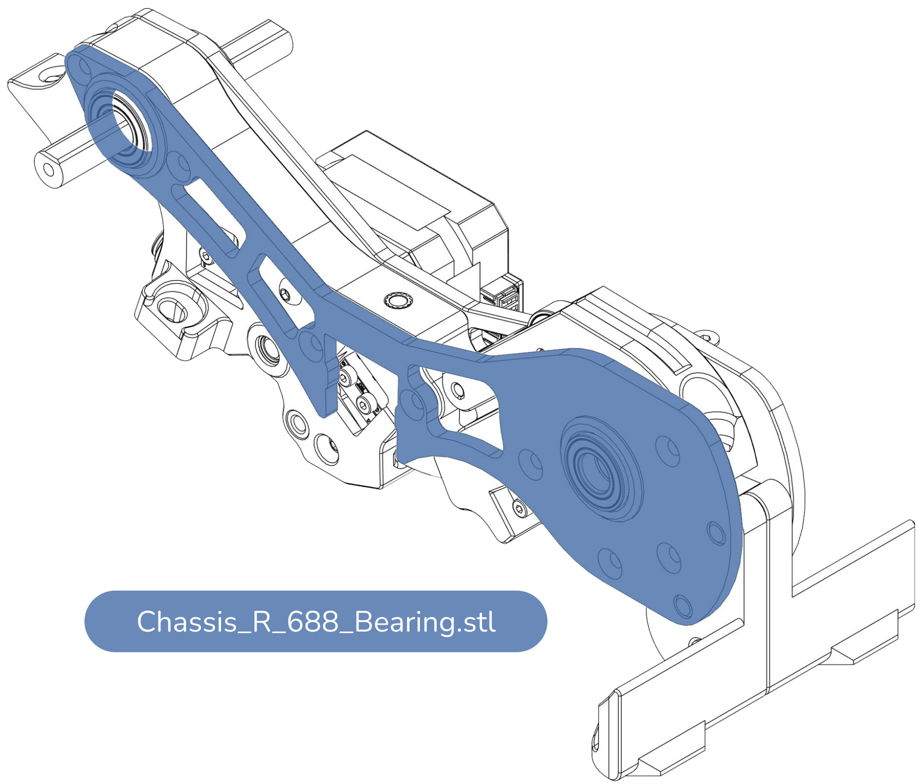
SECURE THE BOX MOUNT SIDES TOGETHER

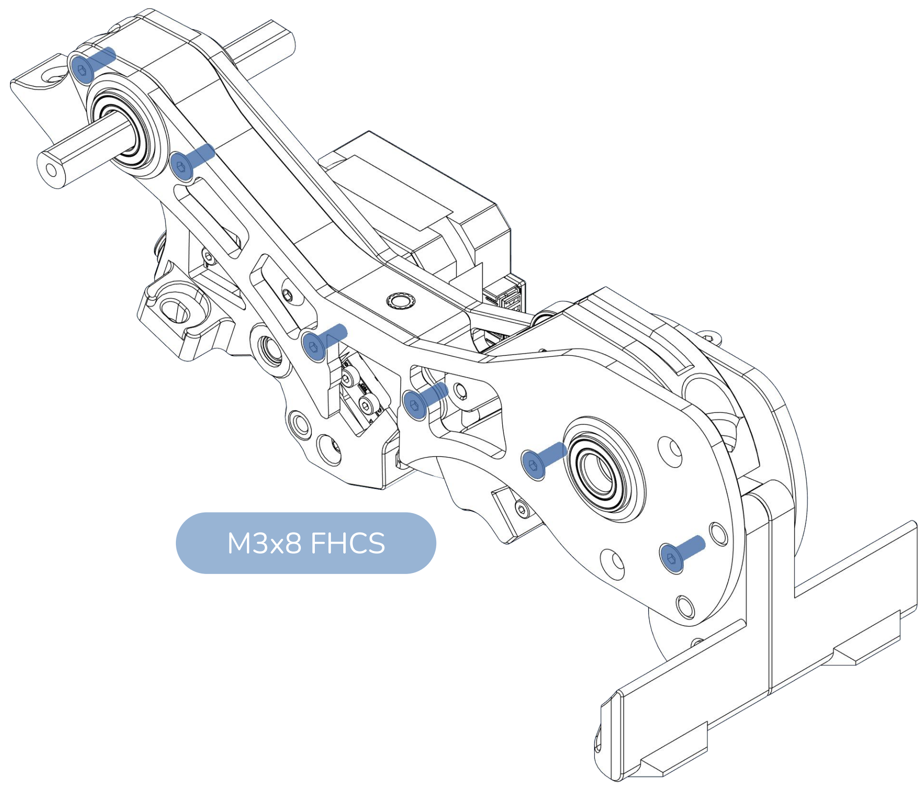


INSTALL THE BOX MOUNT AND IDLE ROLLER AXLE



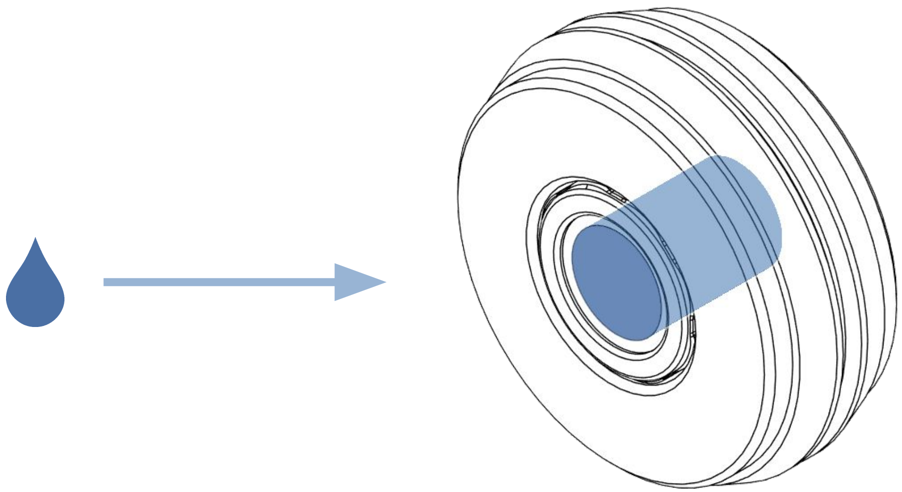
ATTACH THE RIGHT CHASSIS SIDE TO THE ASSEMBLY





Leave the screws about a turn loose. This helps align the shaft with the frame and the bearings.

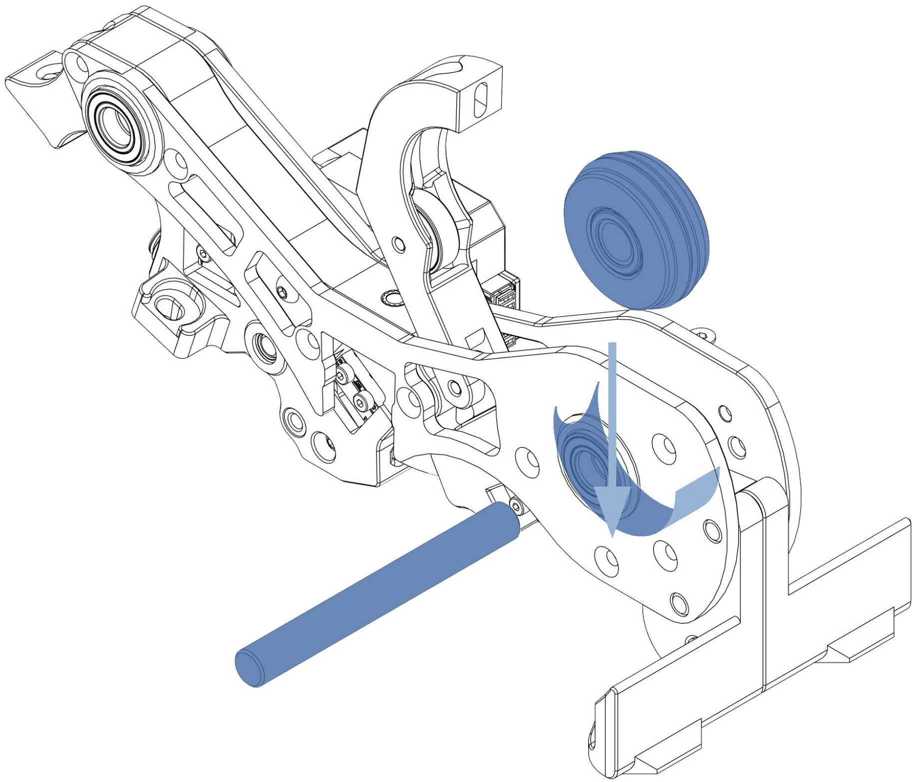
ADD SOME GREASE TO THE CDR ONE WAY BEARING

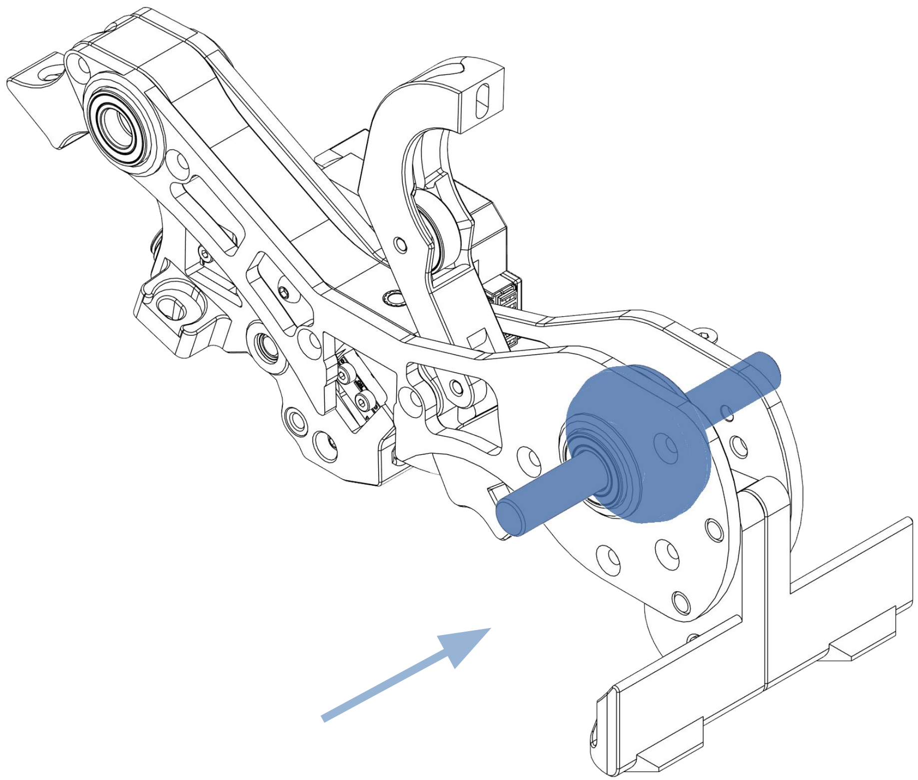


Add a dab of grease to the one way bearing needle rollers and spread it around evenly.

This helps ensure the bearing spins smoothly and releases reliably.

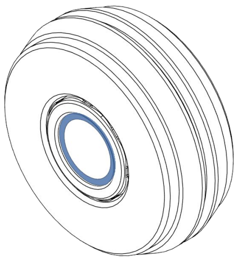
42





If the shaft is too tight against the bearings **use some medium grit sandpaper to sand it down a little.** It needs to be a snug fit.

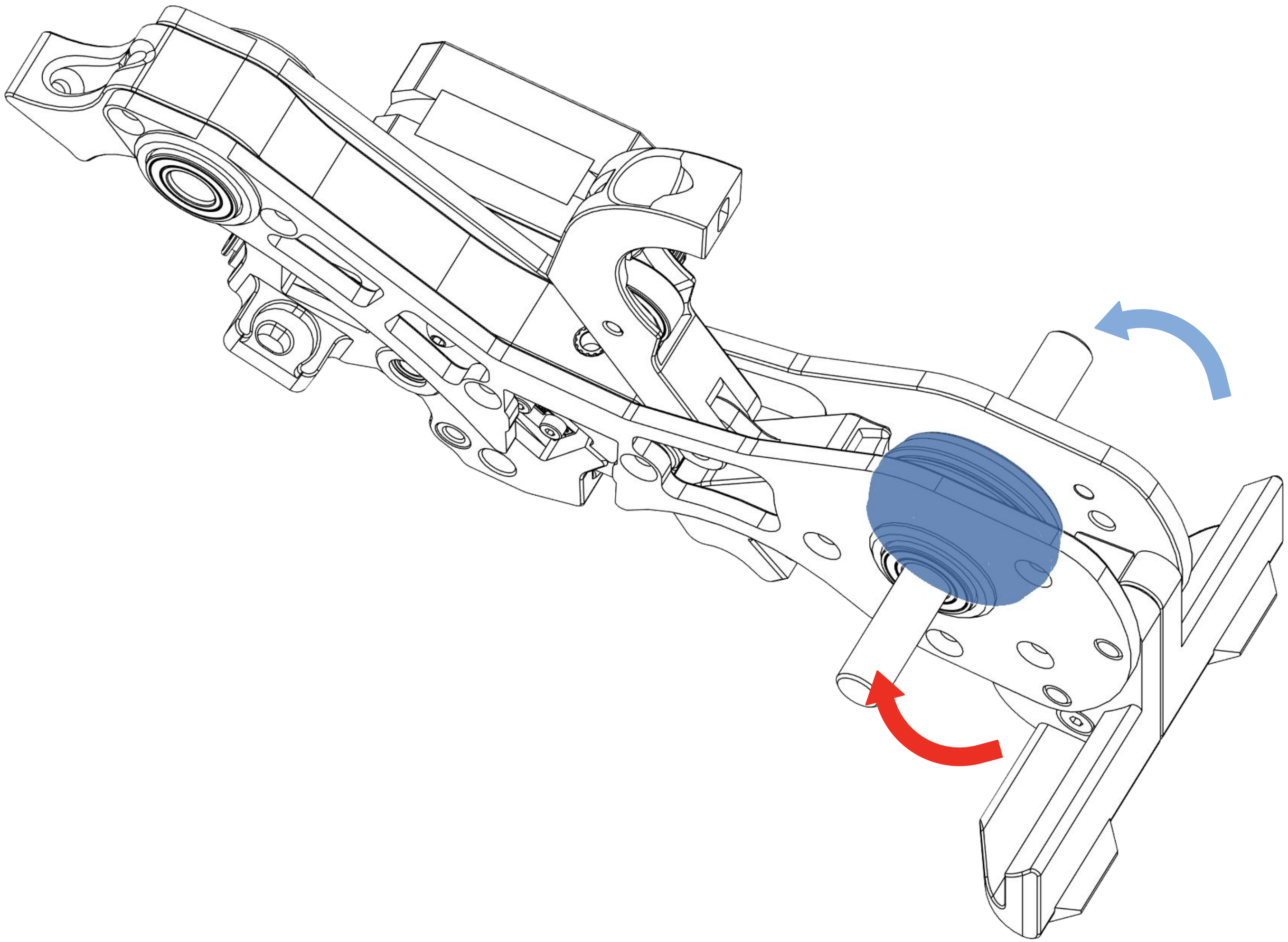
Don't force it in the bearings as then you won't be able to remove the CDR for maintenance!



The highlighted white plastic part should **face towards the left**

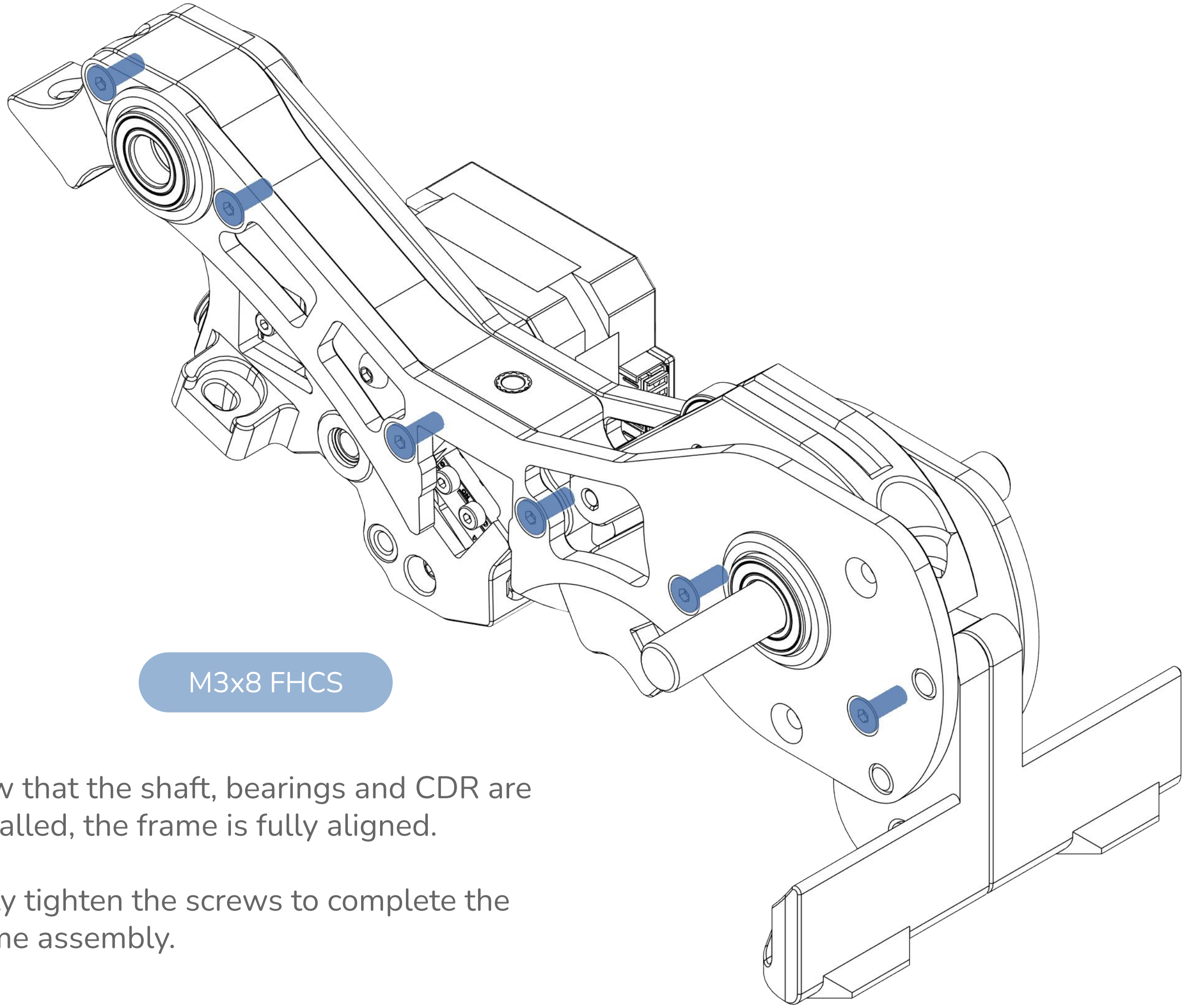
Tip: Use a drill or dremel to spin the shaft while you're sanding it down with the sandpaper.

ENSURE THE ONE WAY BEARING LOCKS & UNLOCKS IN THE RIGHT DIRECTION



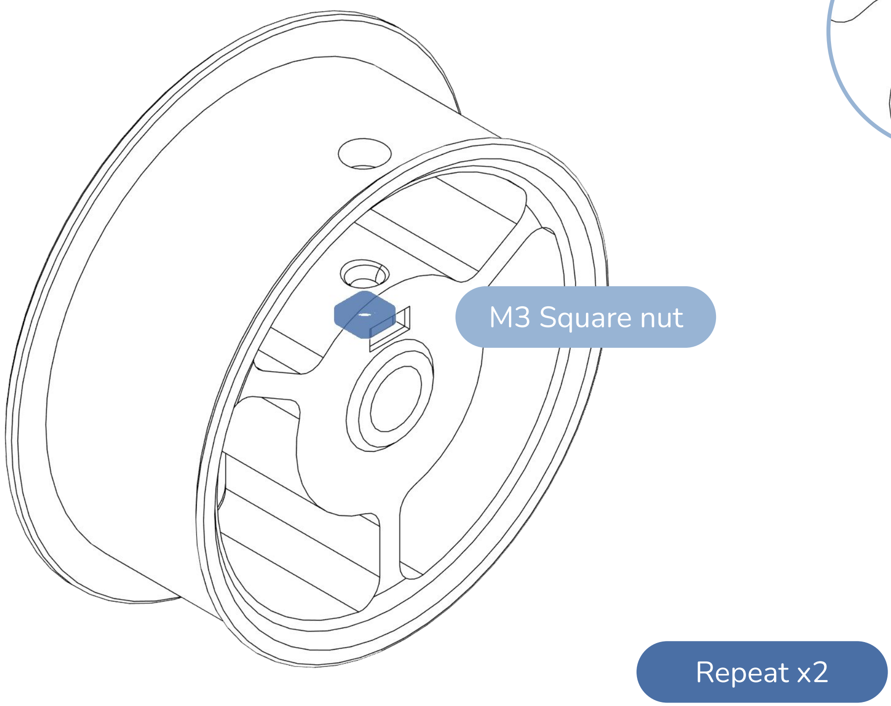
CAUTION:

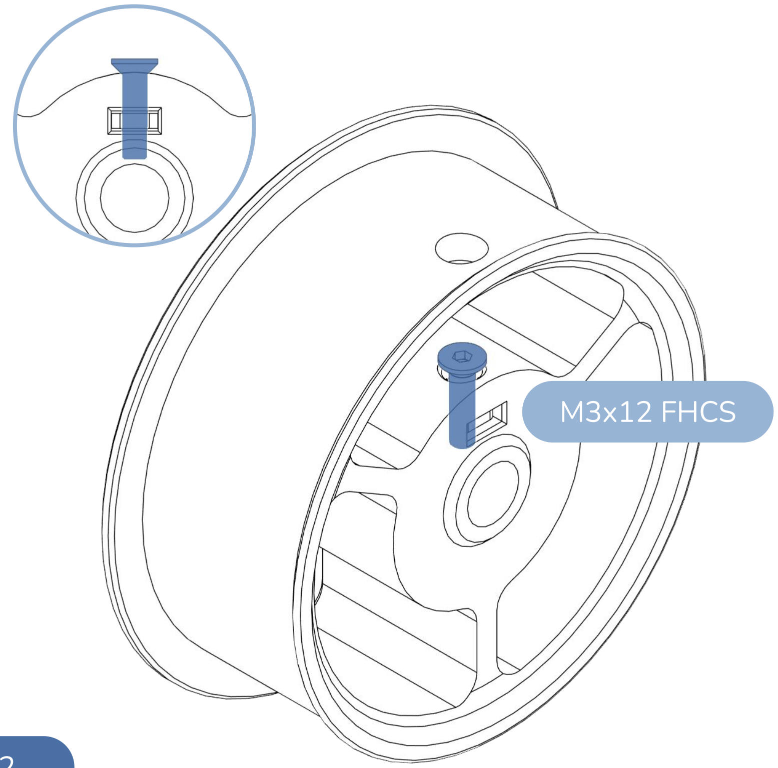
While holding the axle, the CDR should unlock and move freely in the **BLUE** direction, ie pushing filament through the stepper.
It should lock and **not** move in the **RED** arrow direction, ie. pulling filament out.



Now that the shaft, bearings and CDR are installed, the frame is fully aligned.

Fully tighten the screws to complete the frame assembly.

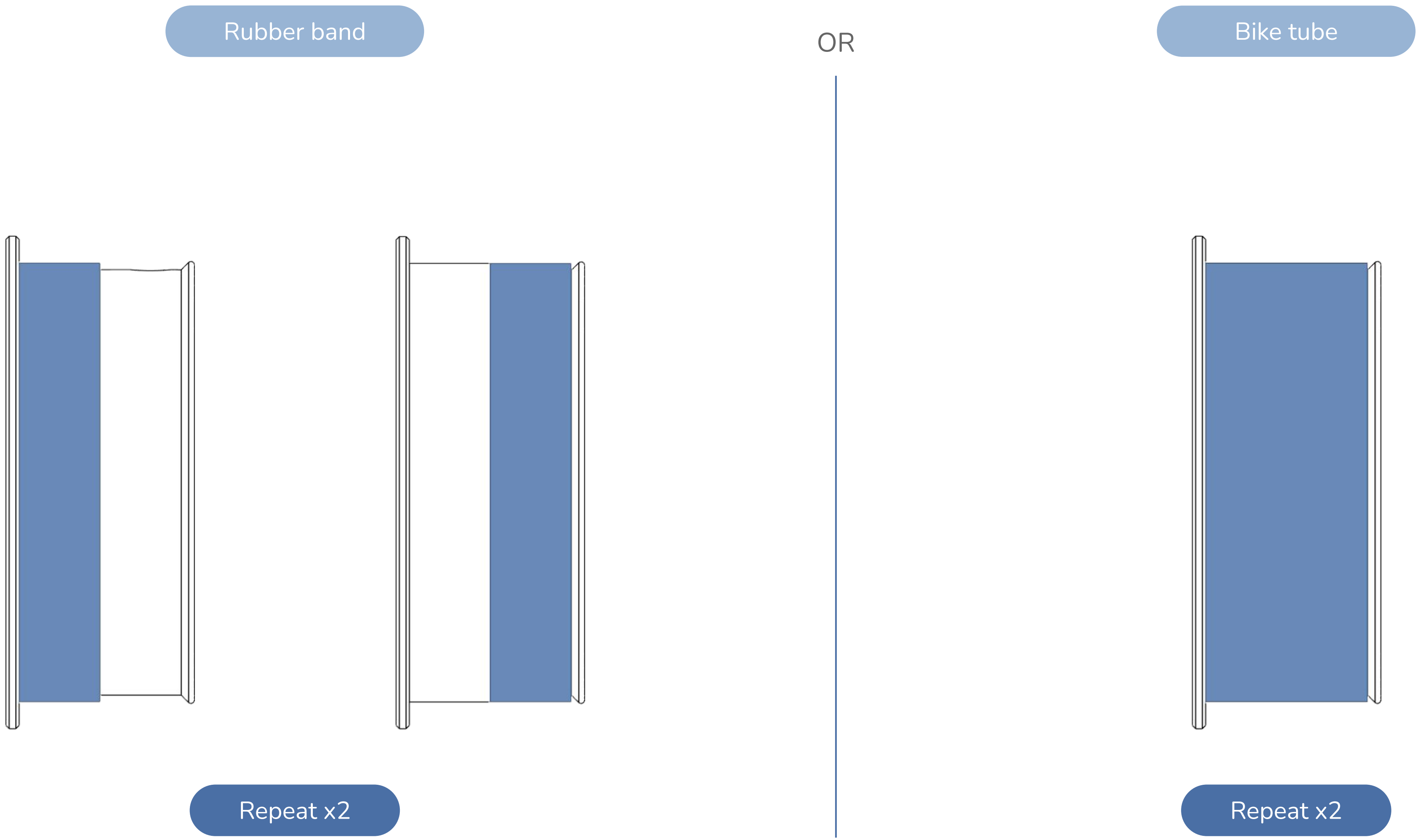




Screw needs to be finger tight and not fully flush with the frame to allow the wheels to be inserted on the driving shaft.

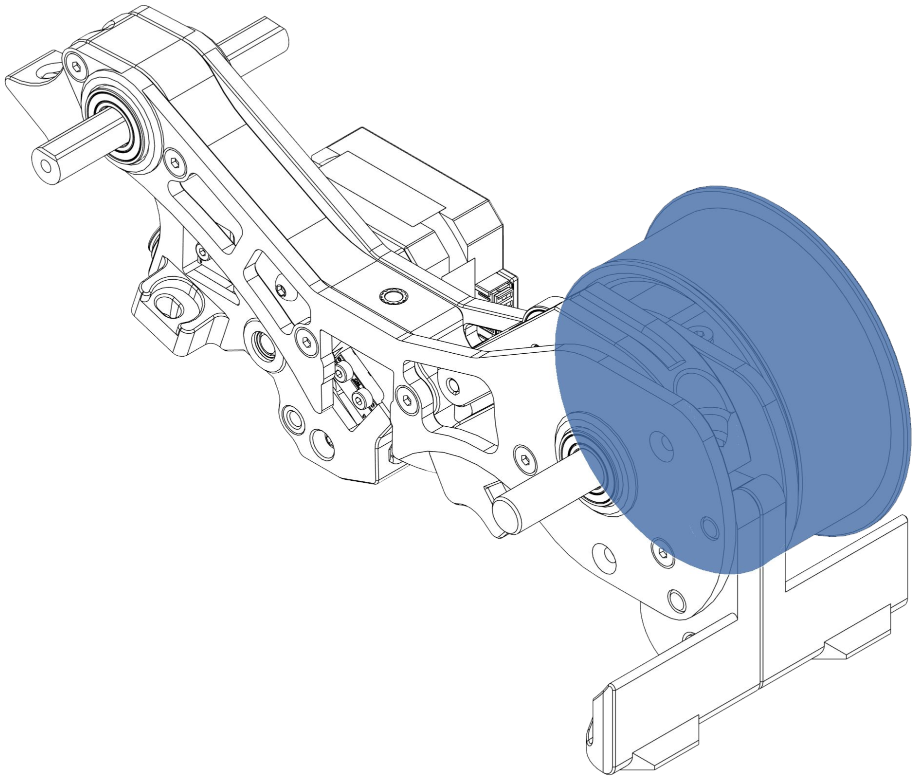
Repeat for the other wheel.

46



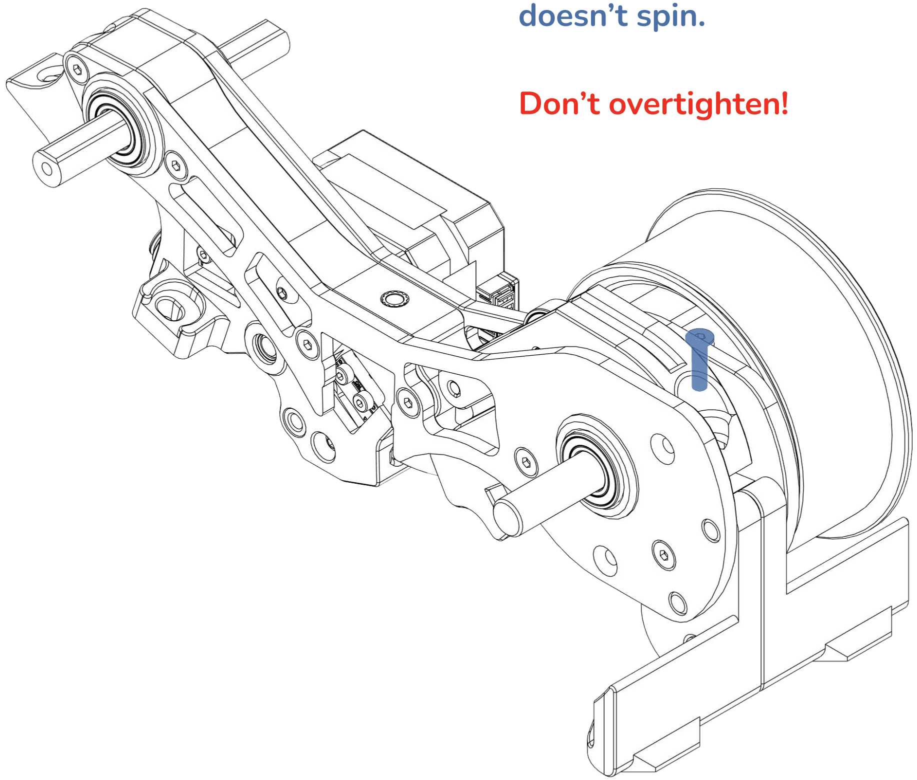
There are two options for creating a "grip surface" on the driving wheels. Rubber bands or a cut bike tube. For ease of assembly rubber bands are recommended.

47

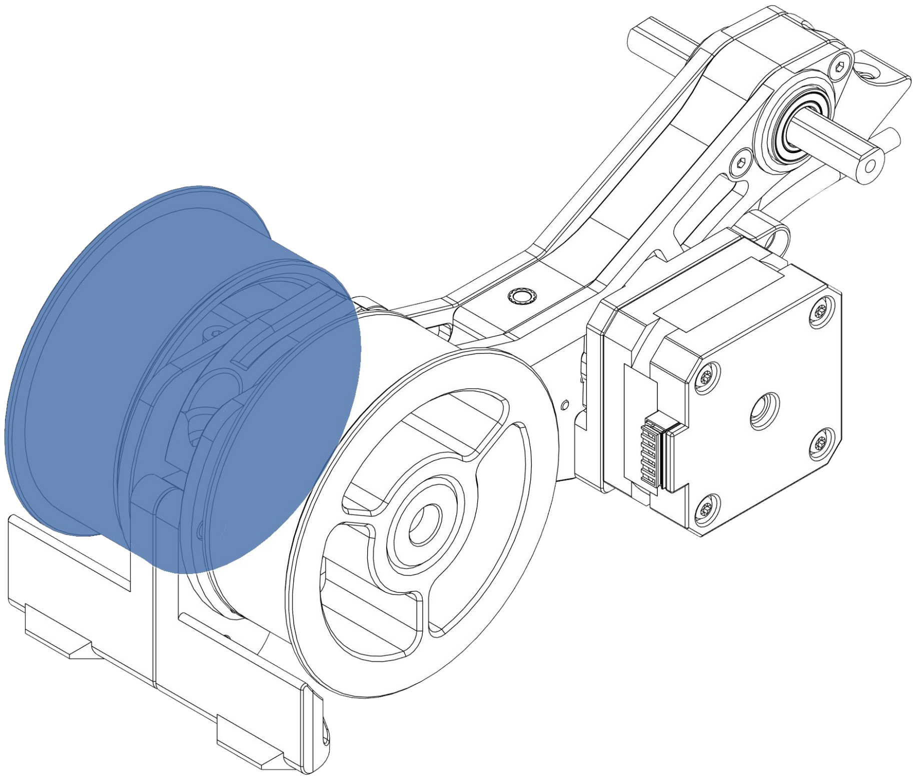


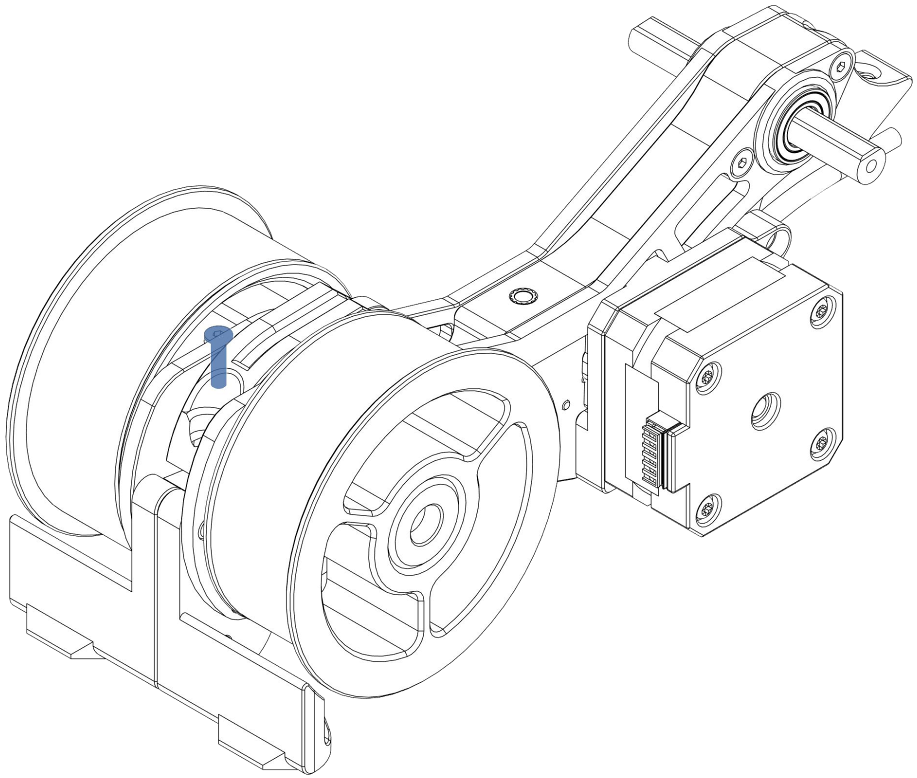
Tighten till the wheel is **snug and doesn’t spin.**

Don’t overtighten!



PLACE AND SECURE WHEEL ON ASSEMBLY

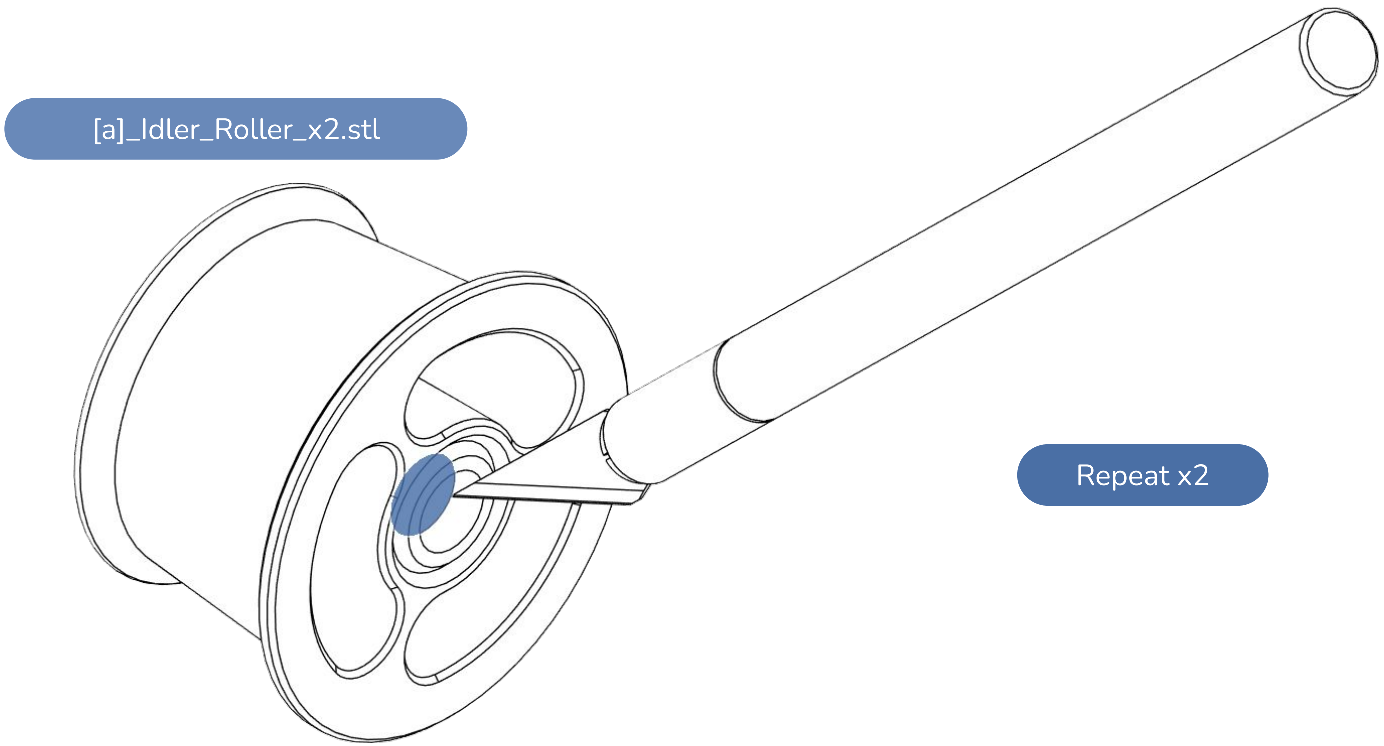




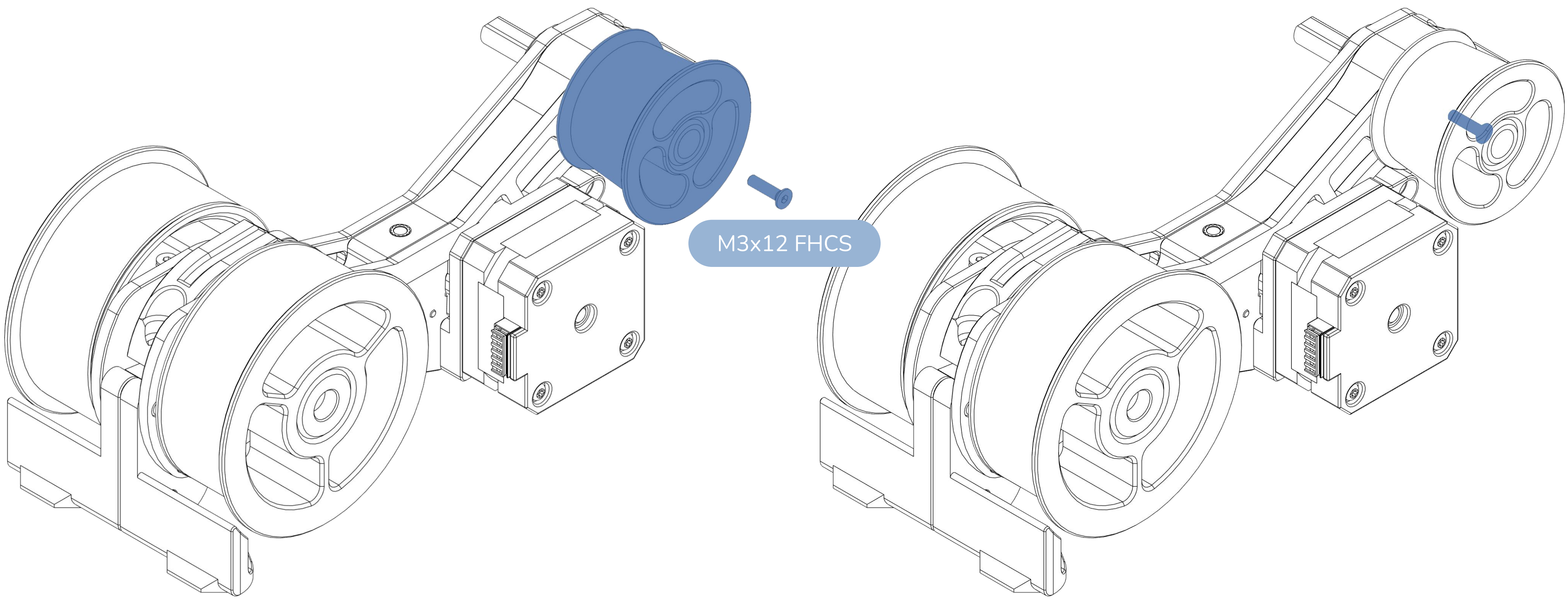
Tighten till the wheel is **snug and doesn't spin.**

Don't overtighten!

CLEAN UP THE SACRIFICIAL BRIDGE ON BOTH IDLER ROLLERS



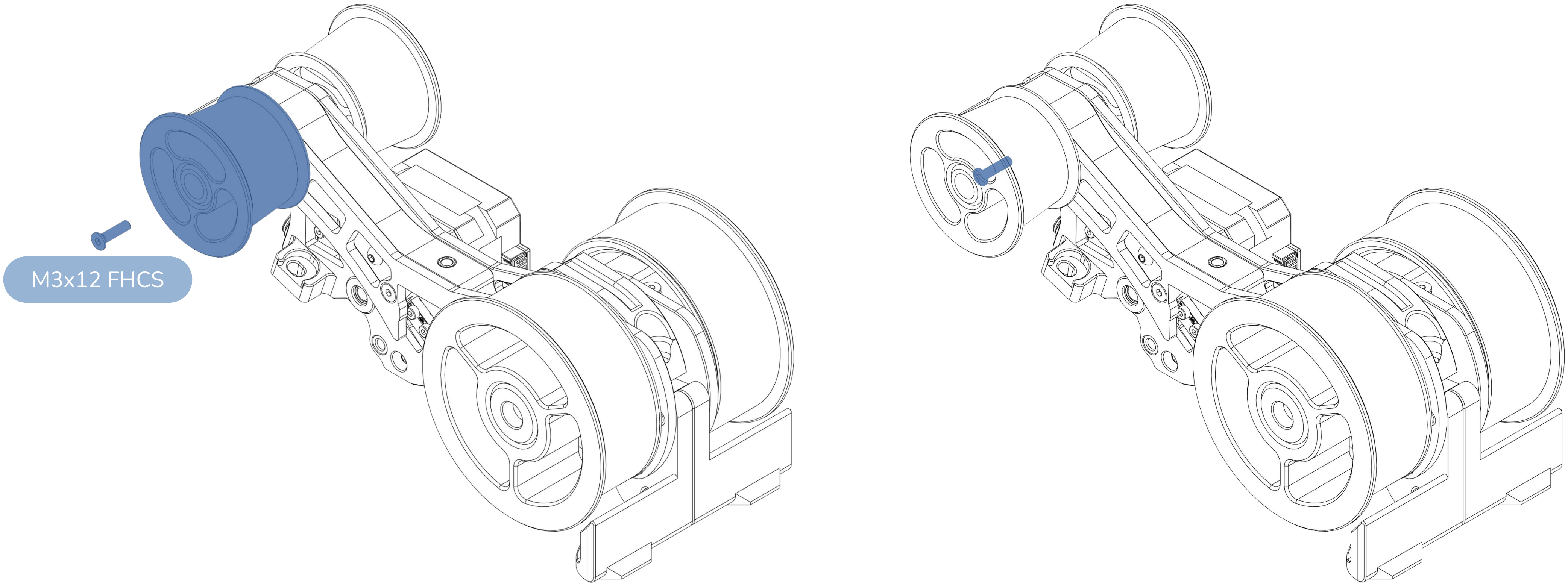
PLACE AND SECURE IDLER ROLLER ON ASSEMBLY



CAUTION:
It is critical that the wheels are able to spin freely without binding and too much friction.

Tighten the M3x12 screw till snug then loosen by half a turn.

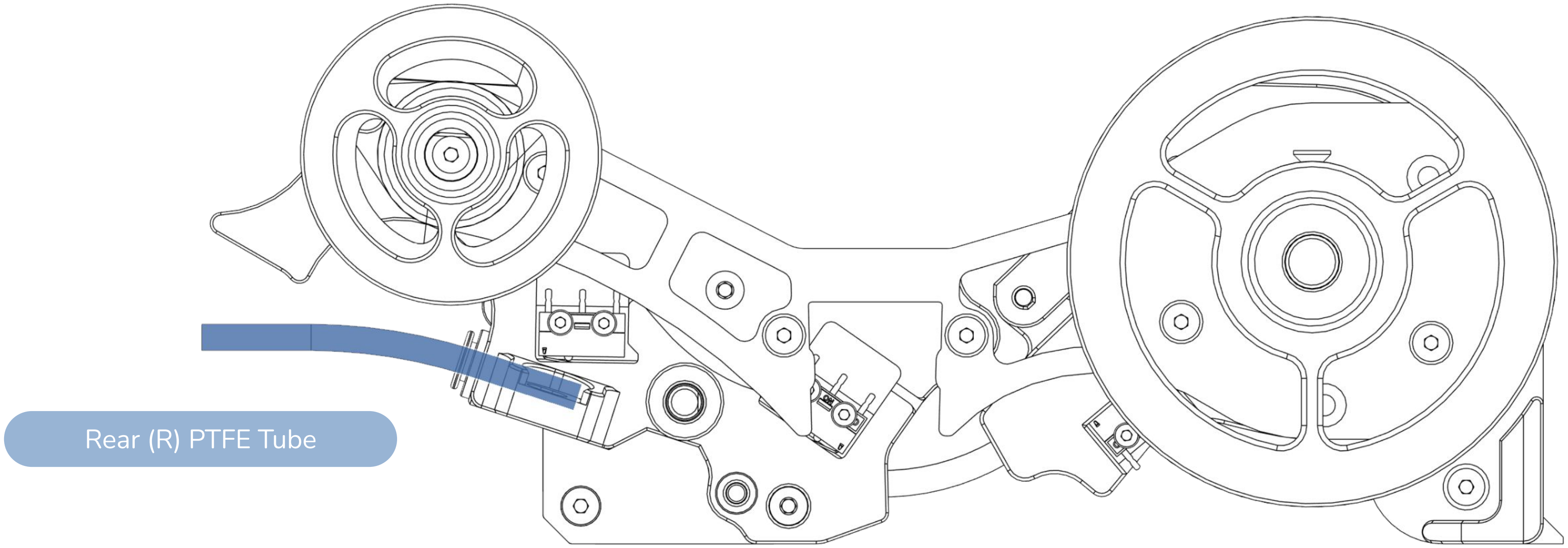
PLACE AND SECURE IDLER ROLLER ON ASSEMBLY



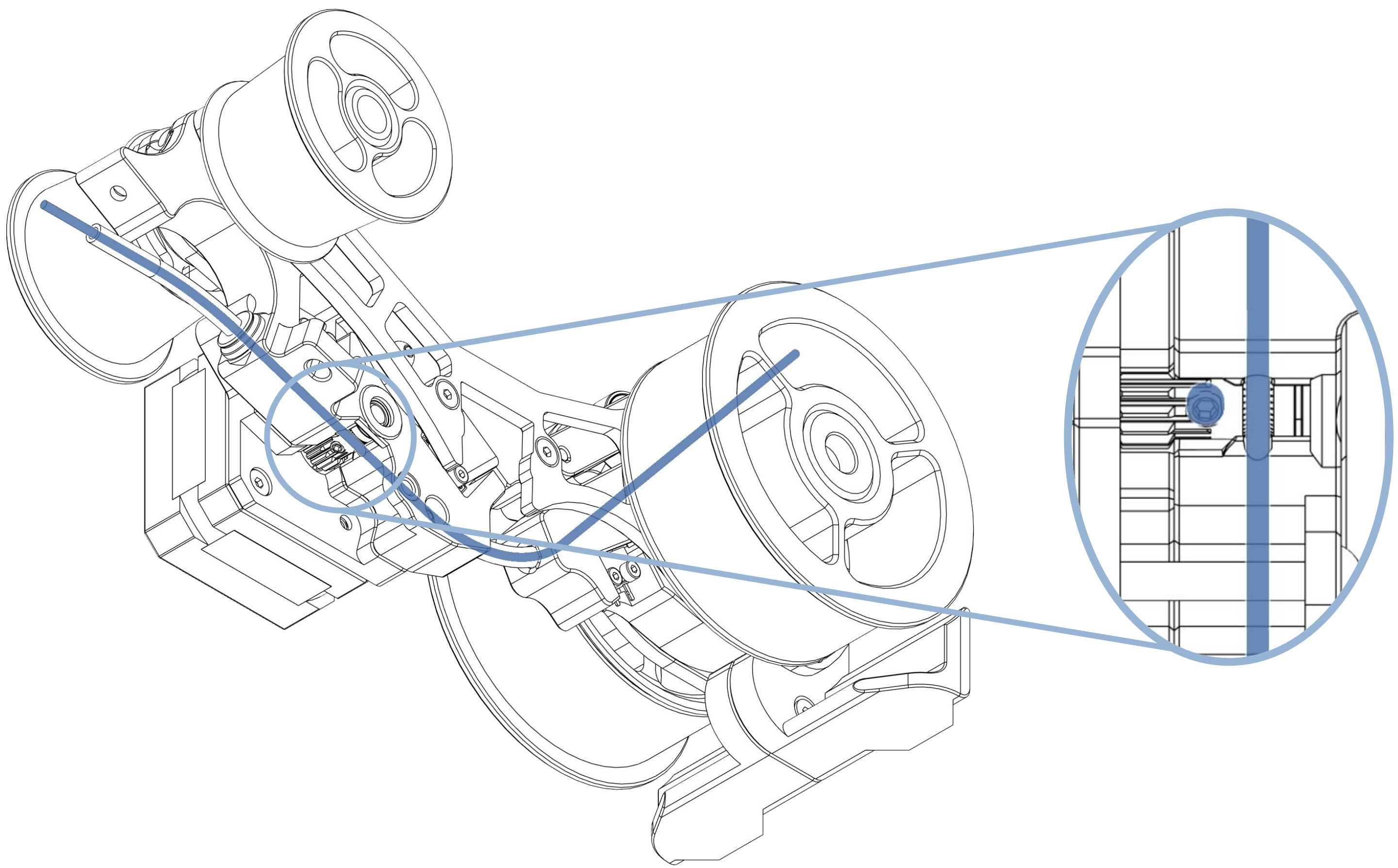
CAUTION:
It is critical that the wheels are able to spin freely without binding and too much friction.

Tighten the M3x12 screw till snug then loosen by half a turn.

INSERT THE REAR PTFE TUBE



ALIGN THE BMG GEAR TO THE FILAMENT PATH



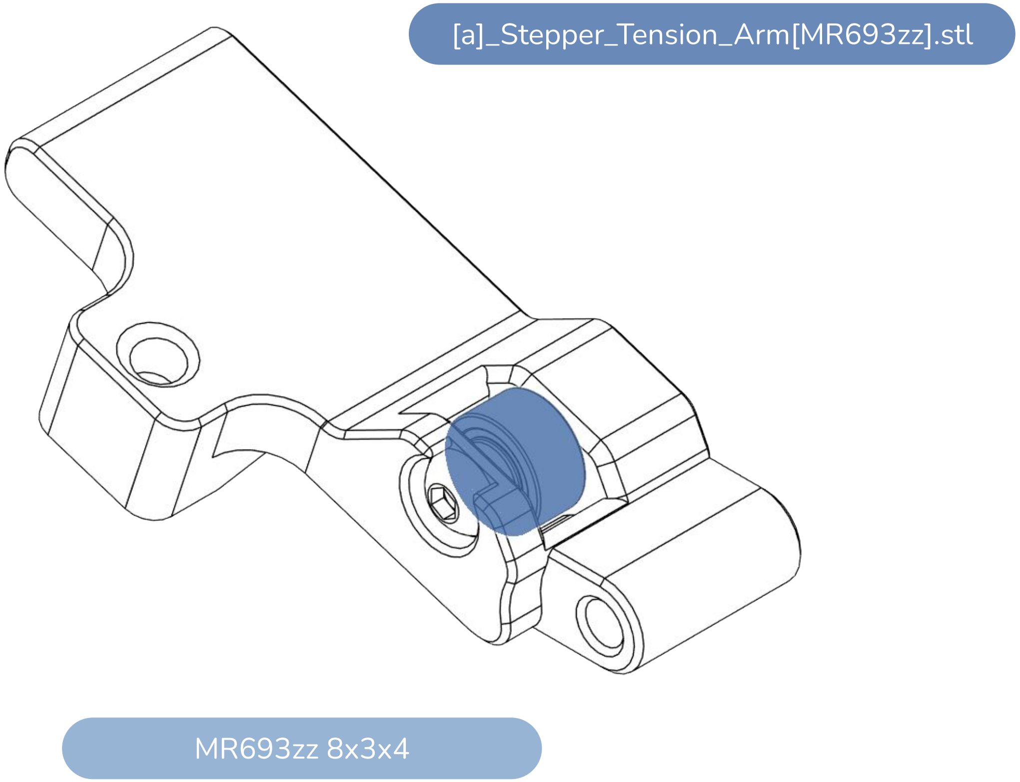
CAUTION:

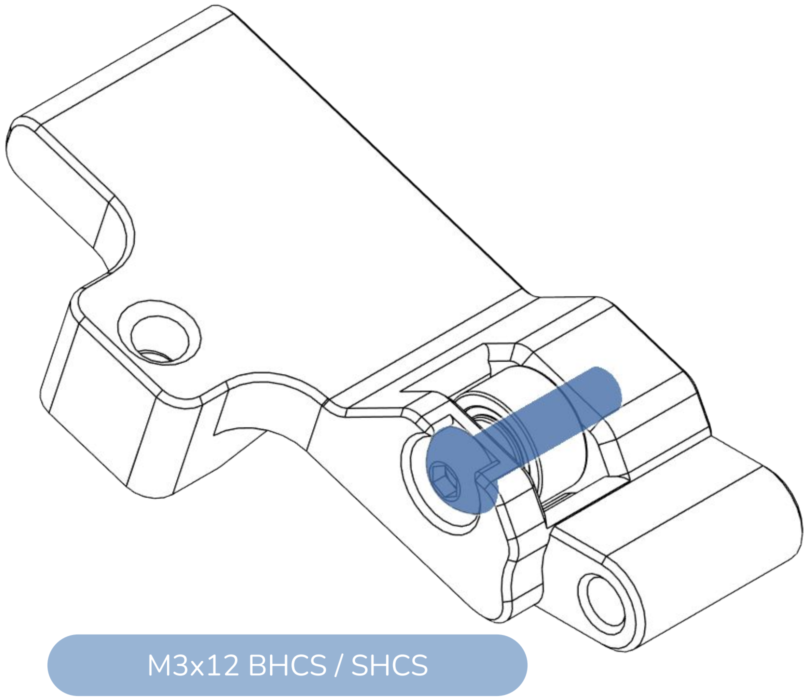
It is critical that the BMG gear is aligned with the filament path and that it does not press against the MR85 bearing. Doing so will increase friction in the system and reduce the unit's reliability.

Add some filament manually to the assembly. Move the filament back and forth till the BMG gear centres with the path. Tighten down. Ensure there is approximately 0.1-0.2mm gap between the BMG gear and the bearing.

54

INSERT & SECURE TENSION ARM BEARING

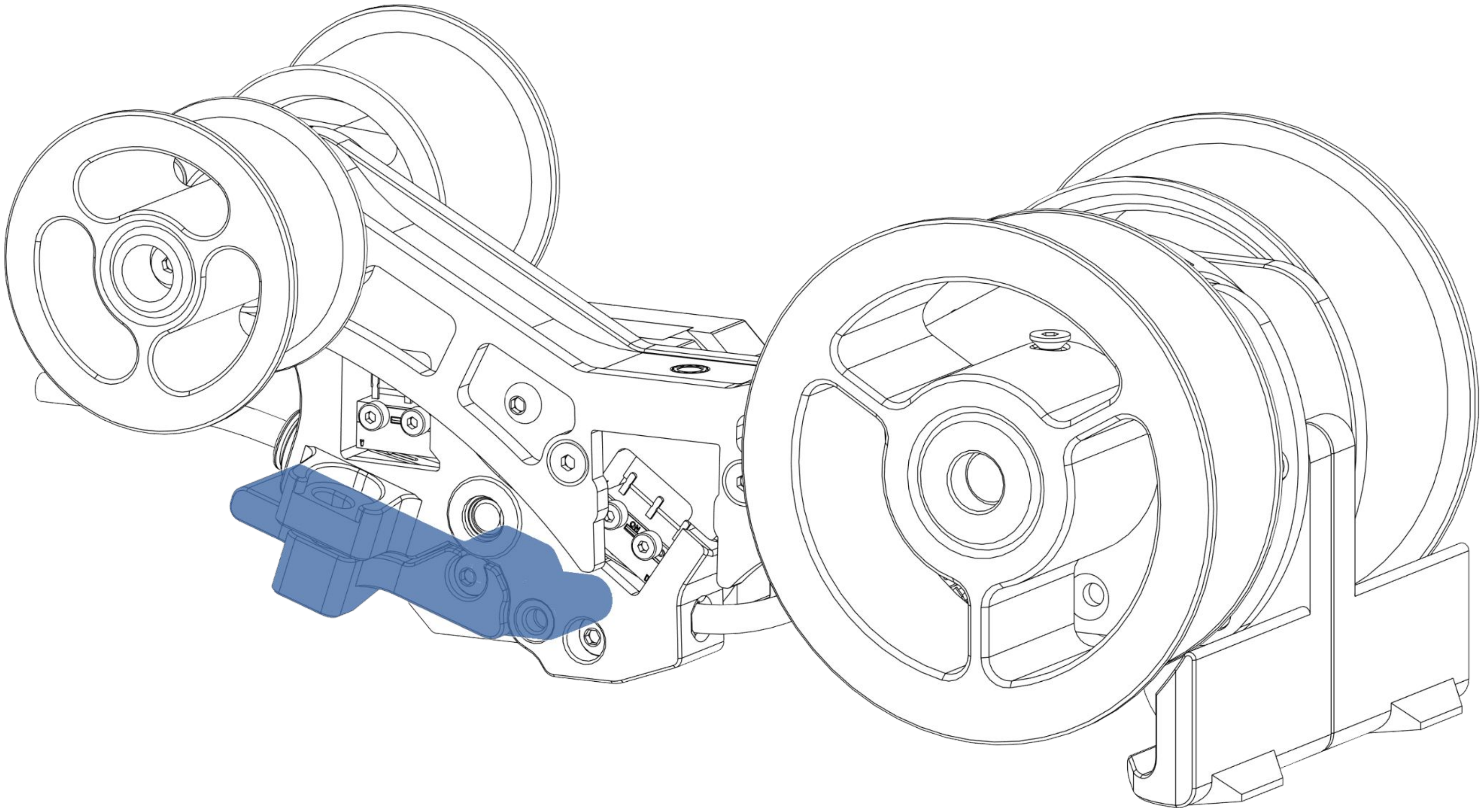




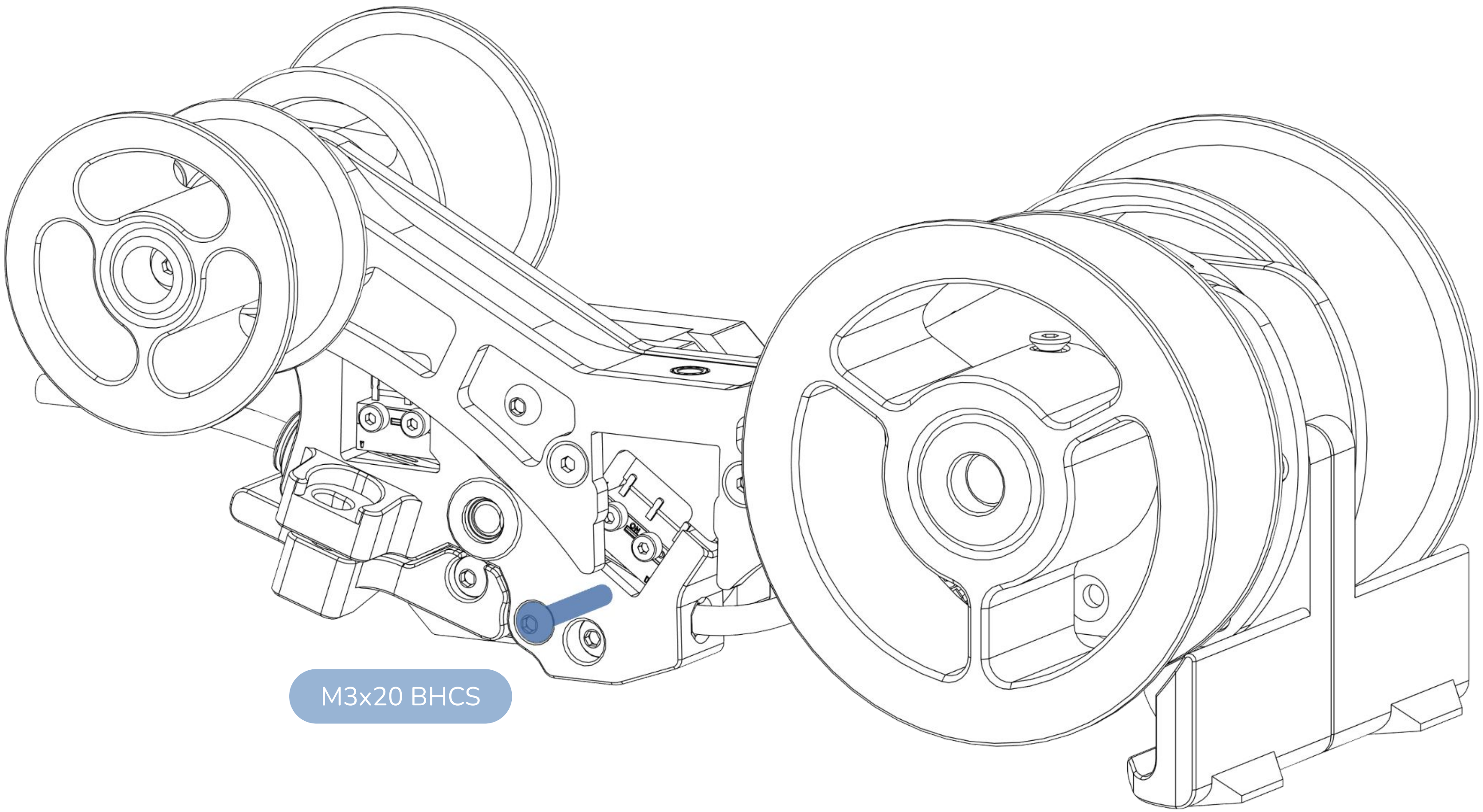
CAUTION:
It is critical that the BMG idler bearing is able to spin freely with little/no friction and no binding.

Tighten the M3x12 screw till snug then loosen by half a turn. The bearing may be able to wiggle in the assembly. That is perfectly fine.

PLACE THE TENSIONING ARM INTO POSITION



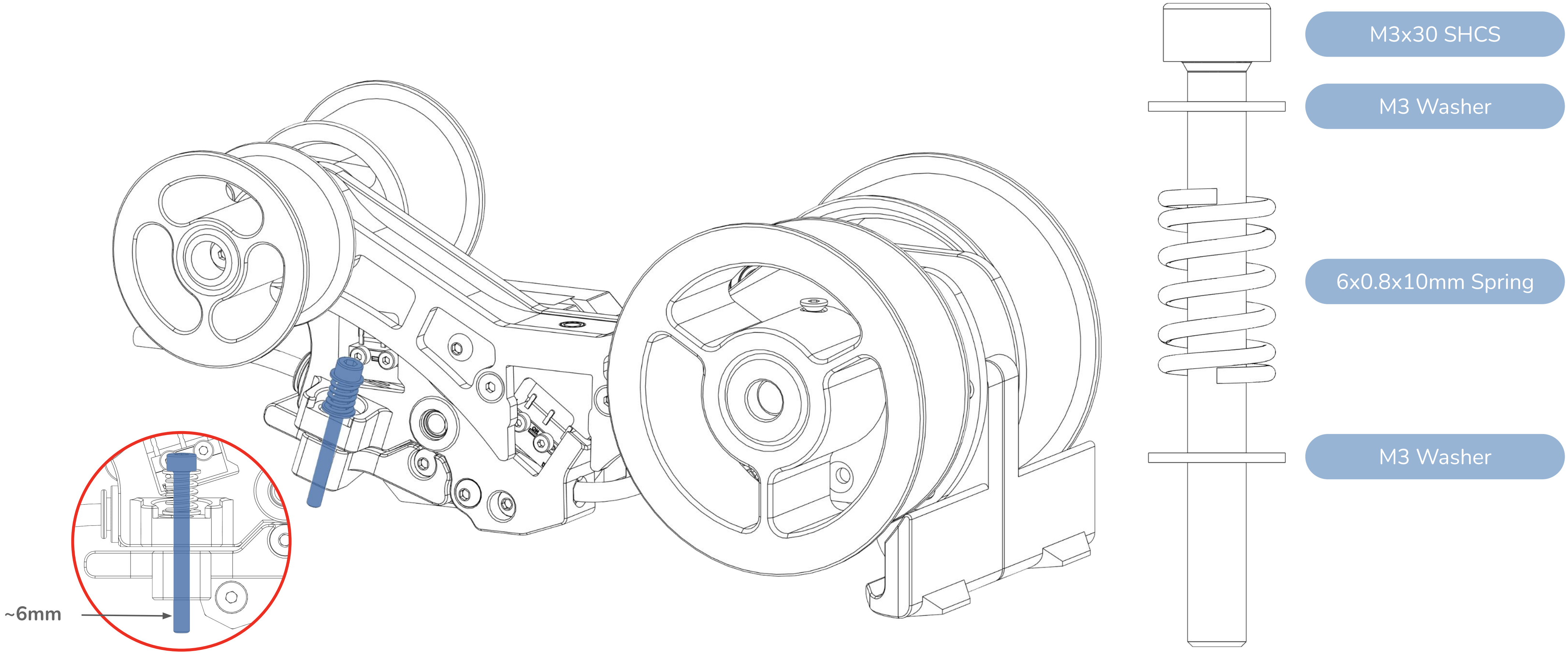
SECURE THE TENSIONING ARM



CAUTION:

It is critical that the tensioning arm is able to move with little/no friction and no binding. It should be able to open under its own weight.

Tighten the M3x20 screw till snug then loosen by half a turn. The arm may be able to wiggle in the assembly. That is perfectly fine.

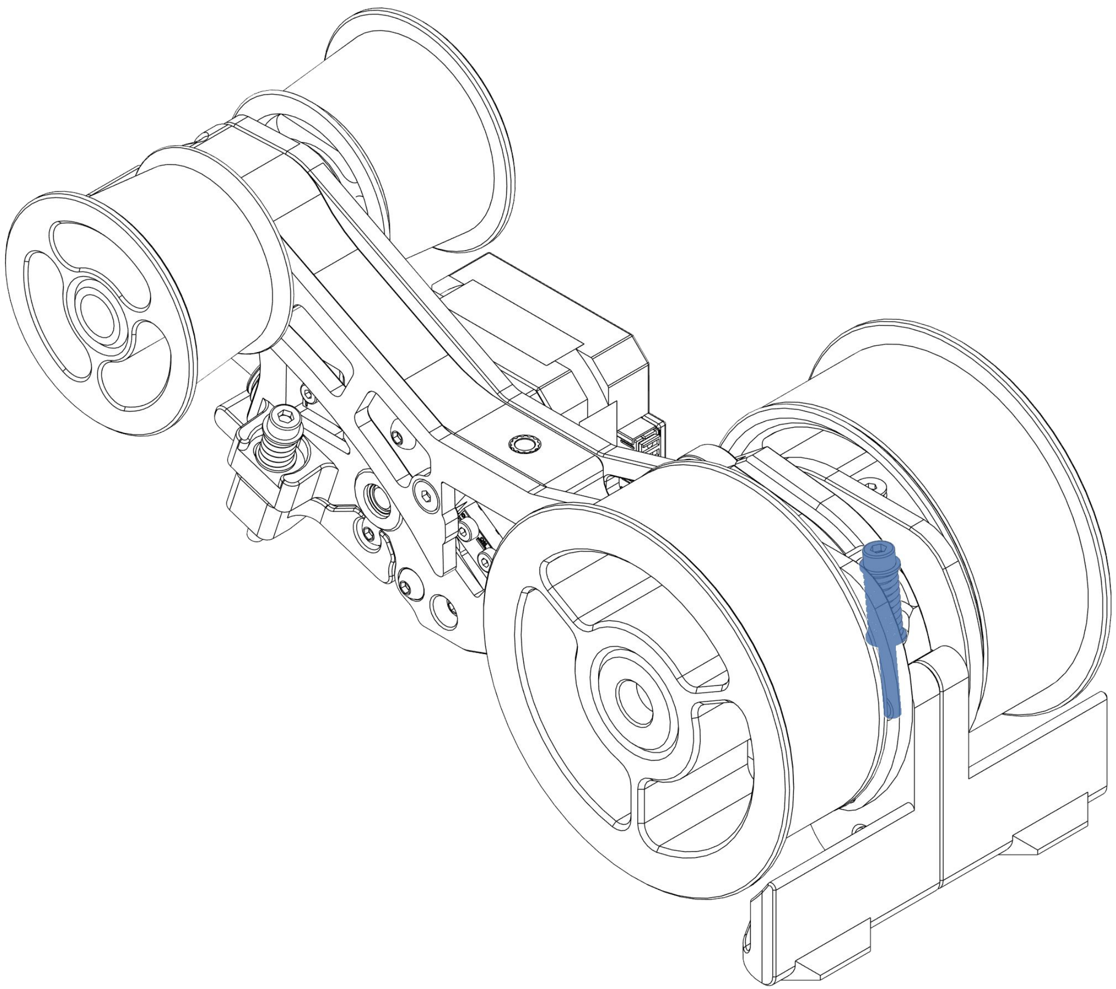


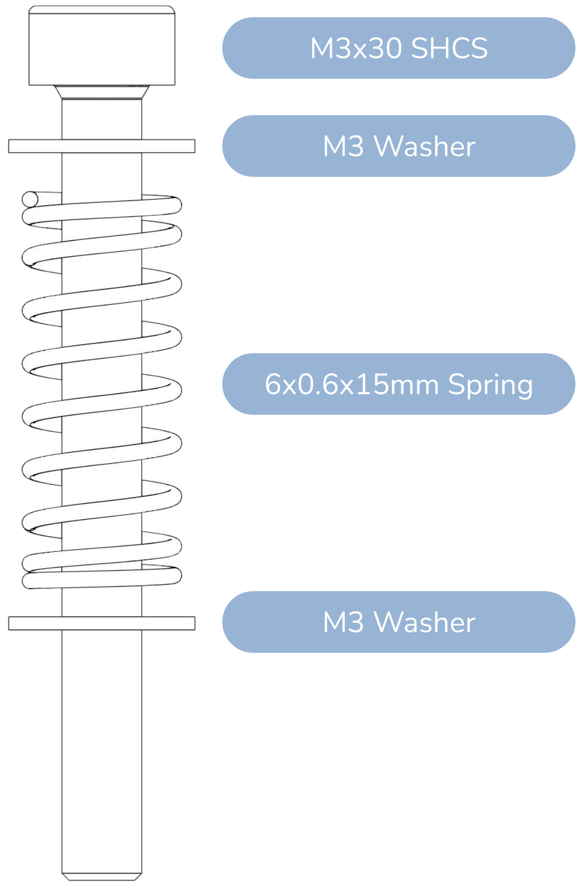
CAUTION:
It is critical that the right amount of tension is placed on the filament. Too little and the filament will slip and grind. Too much and loading speeds will be impacted.

Tighten the M3x30 screw till ~6mm of thread protrudes from the bottom side of the tensioning arm.

Move the tensioning arm by hand. The screw threads shouldn't rub on the stepper body. If they do, remove the screw and adjust the angle of the heatset.

ADD FRONT TENSIONER SCREW

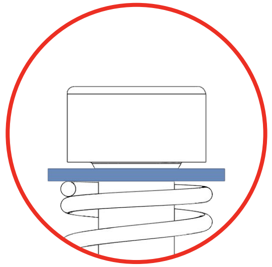




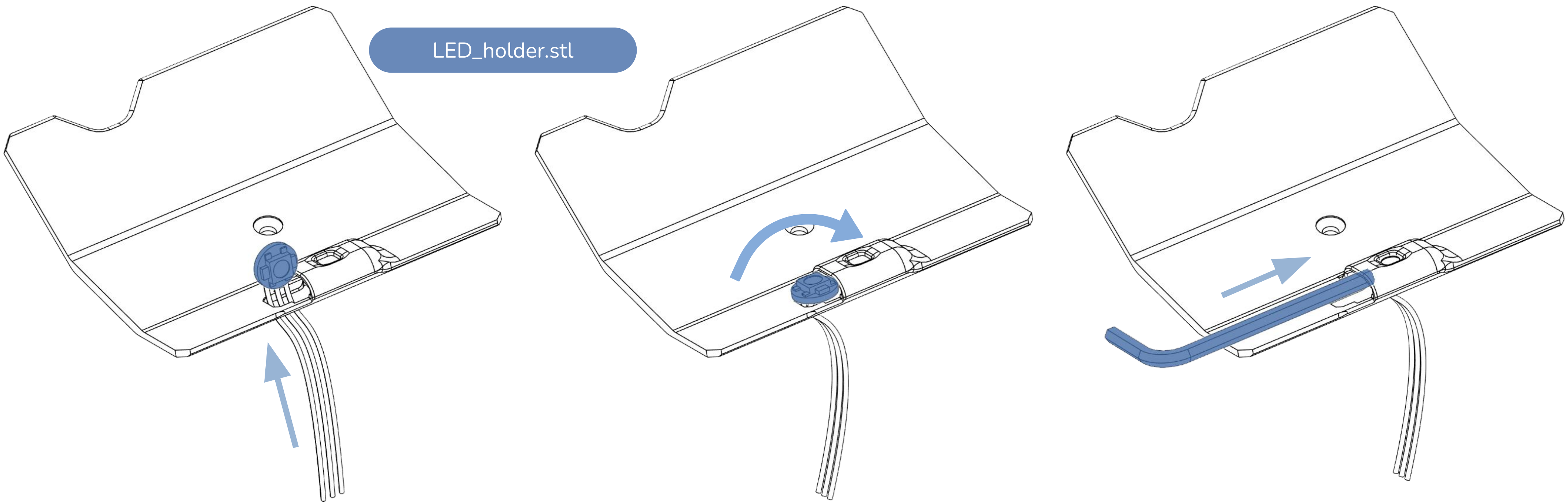
CAUTION:

It is critical that the right amount of tension is placed on the filament by the CDR. Too little and the spool will not rewind tightly. Too much and the filament may break at the CDR.

Without any filament loaded, tighten the M3x30 screw till the top washer barely touches the screw head. The screw may feel loose. That is perfectly fine.



INSERT THE NEOPIXEL LED INTO ITS HOLDER

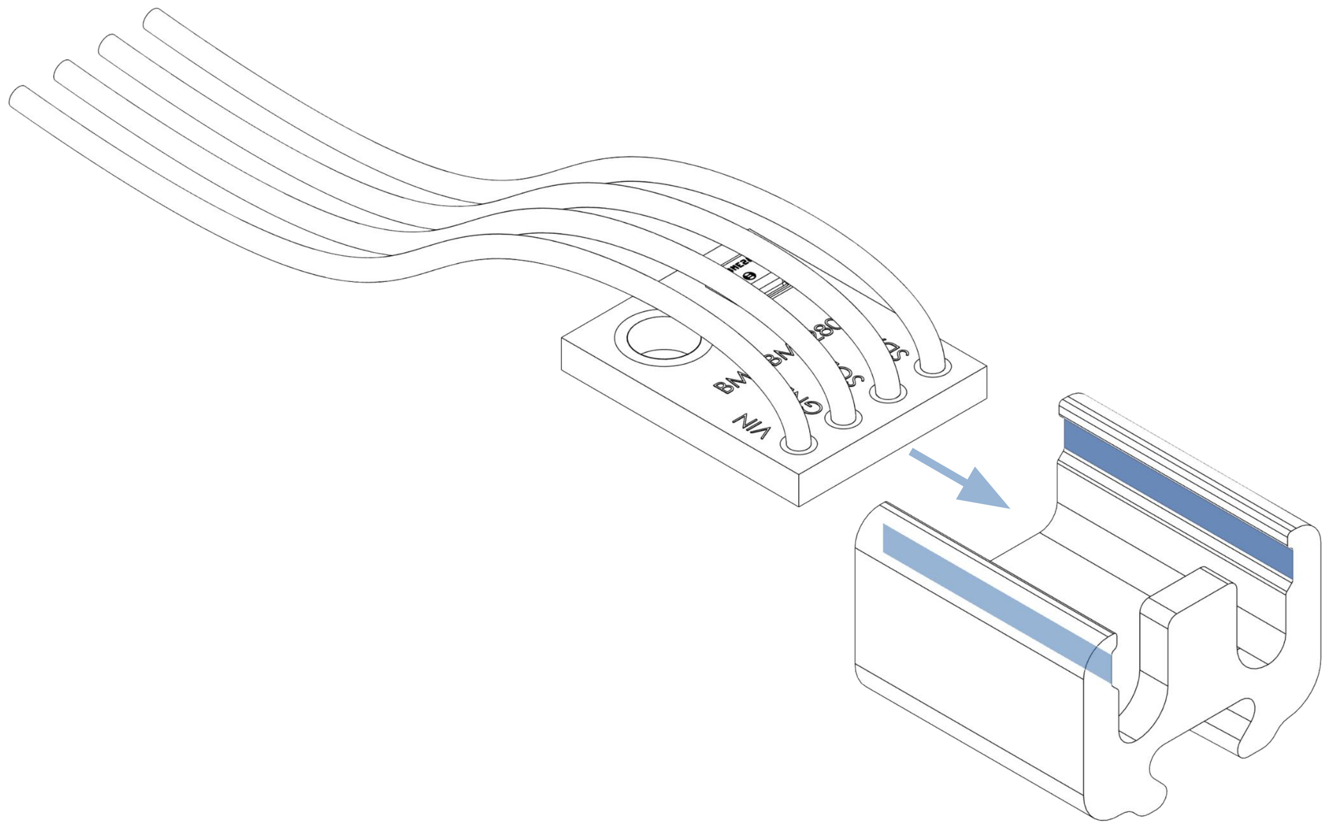


Step 1: Pass led through cover hole

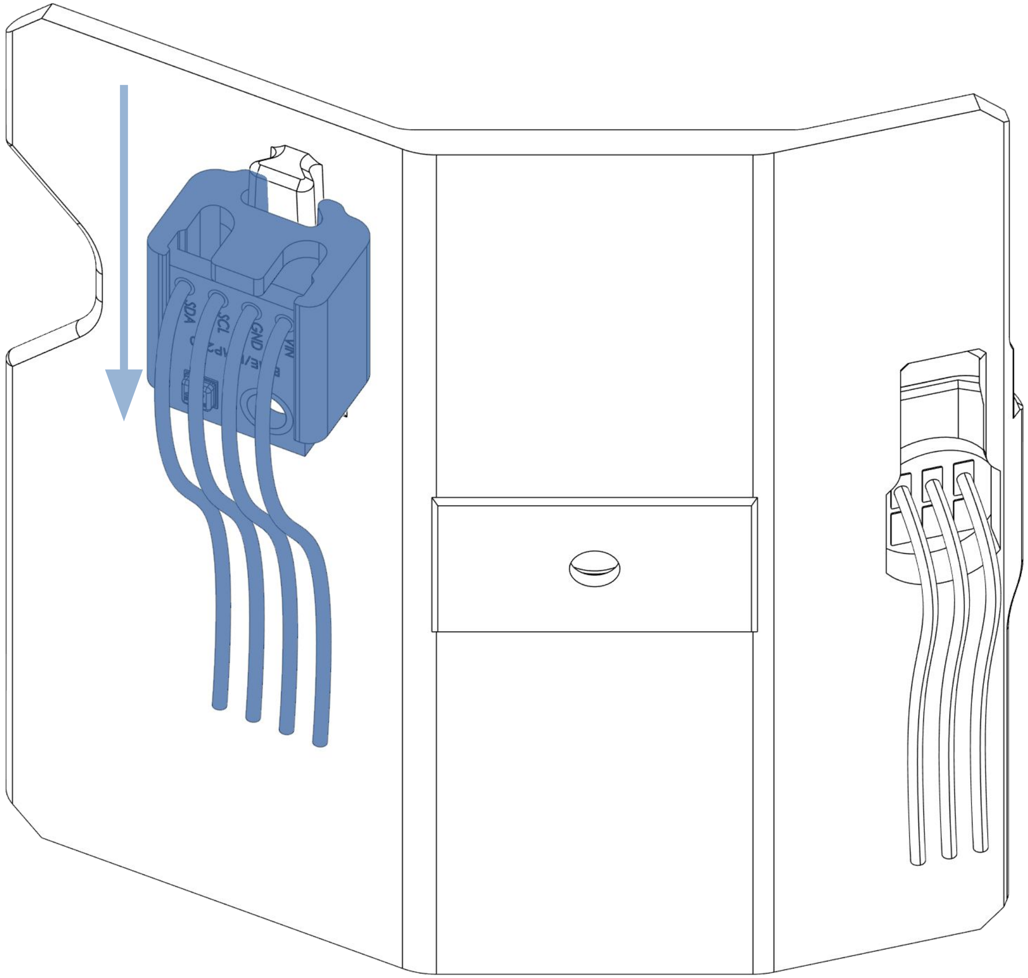
Step 2: Rotate led into position

Step 3: Push led into position using a hex key

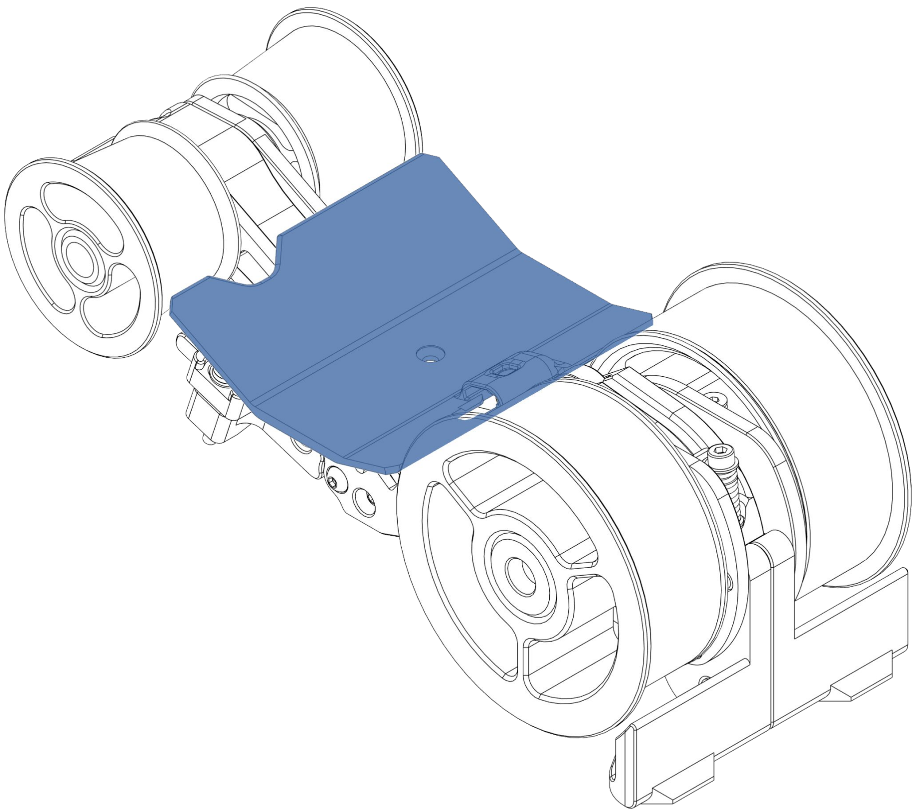
INSERT THE BME280 / ATH20 SENSOR INTO ITS HOLDER

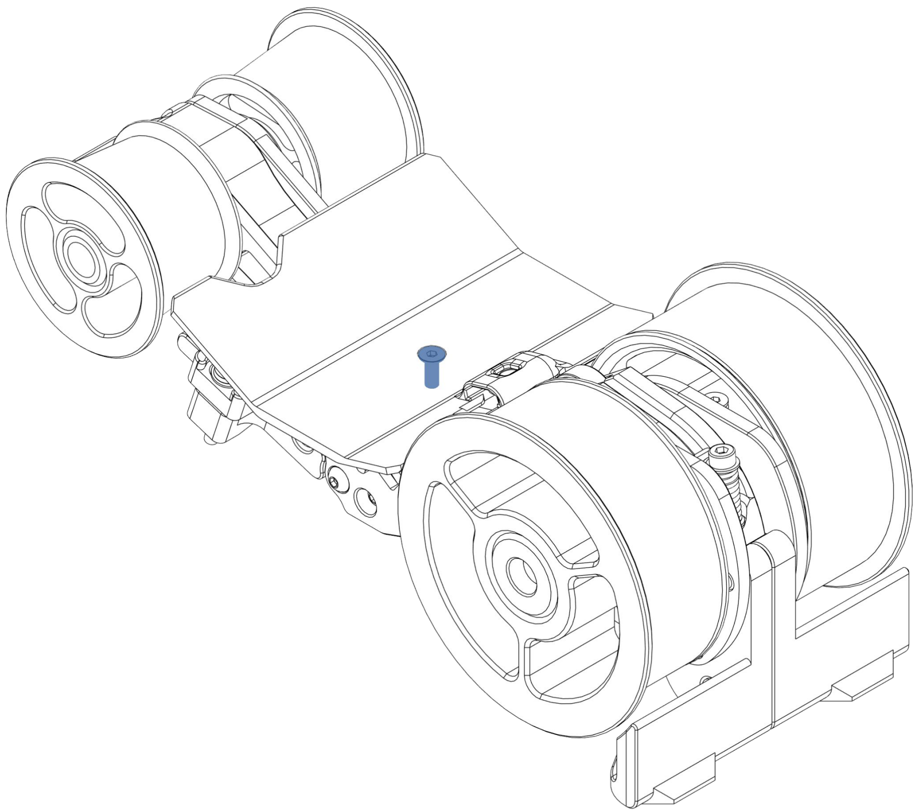


[a]_BME280_Holder.stl


SDA
SCL
GND
VIN

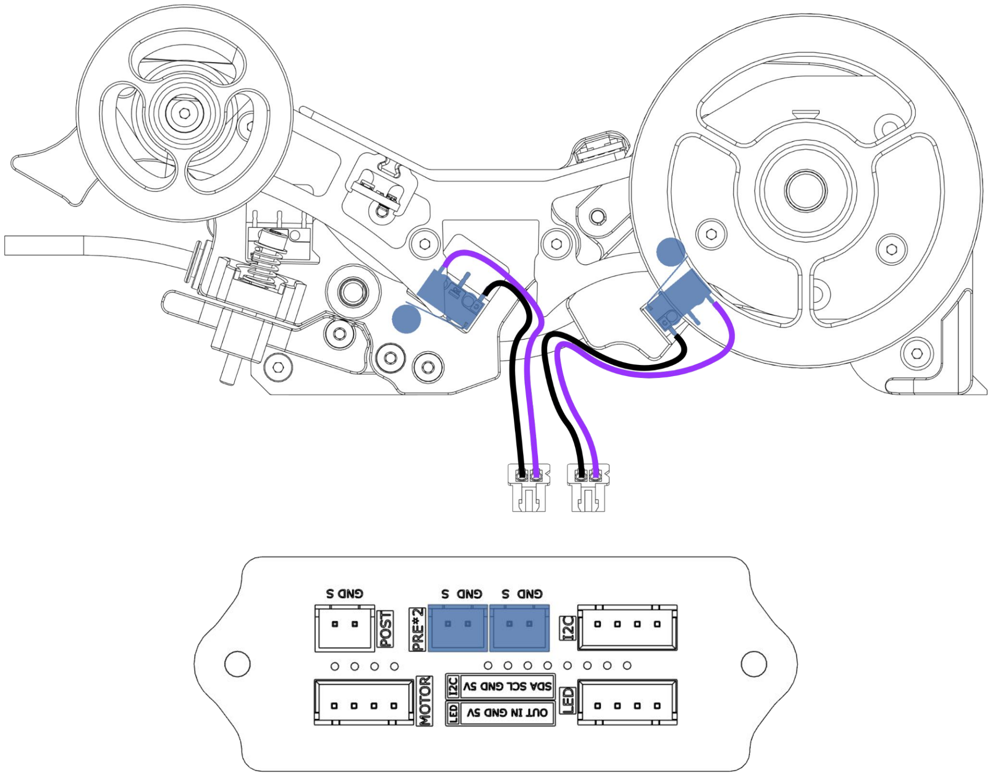
PLACE & SECURE THE LED HOLDER



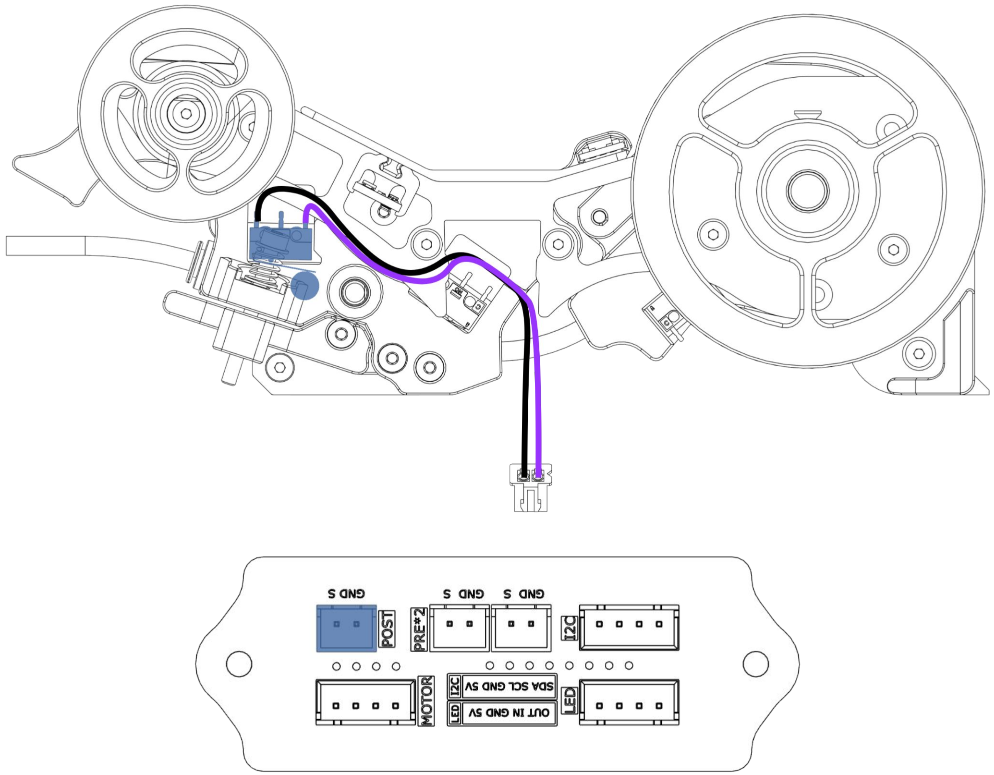


CHAPTER 3: CABLE MANAGEMENT

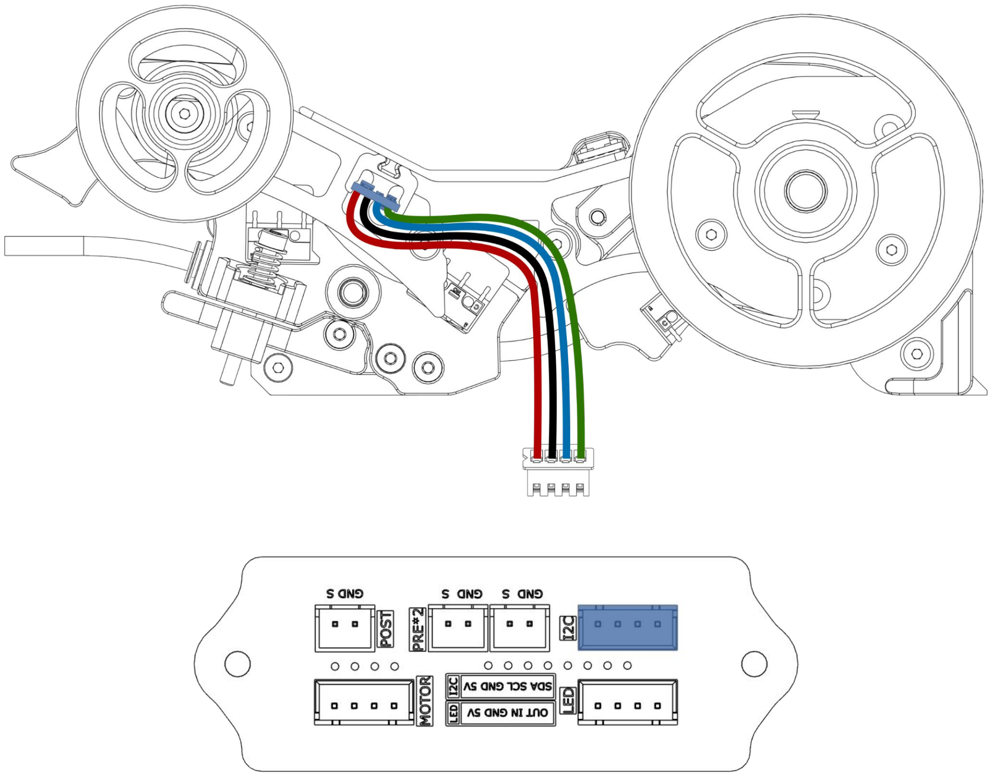
Connect the entry and pre-stepper sensors to the hatch board



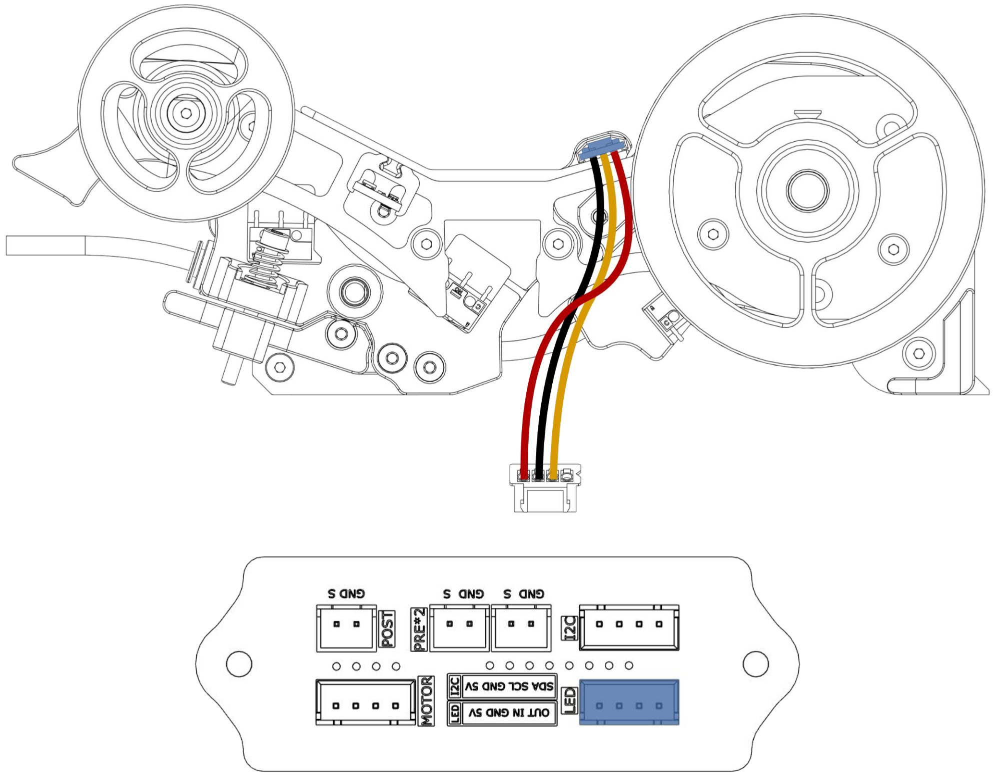
Connect the post stepper sensor to the hatch board



Connect the environment sensor to the hatch board



Connect the neopixel to the hatch board

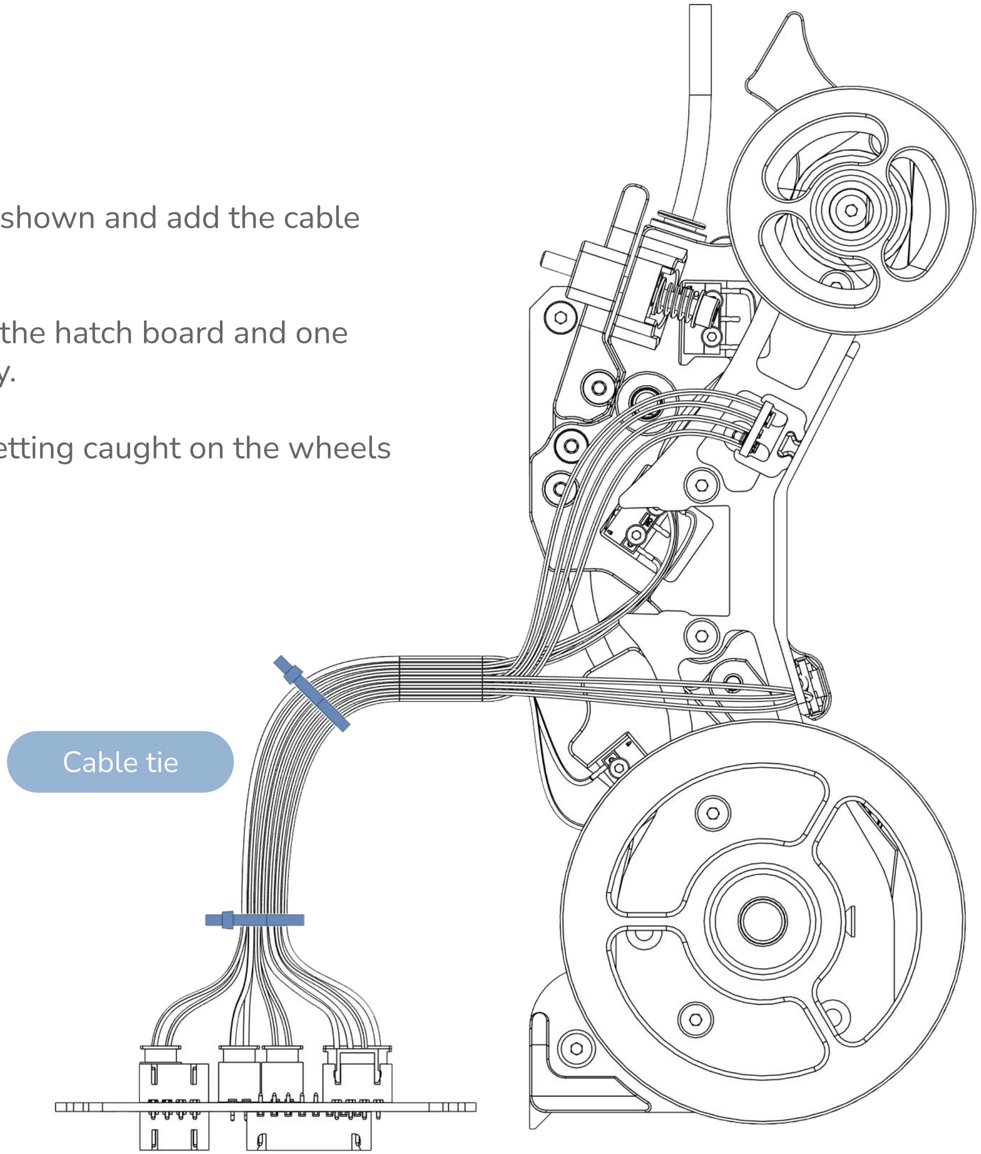


SECURE THE WIRES

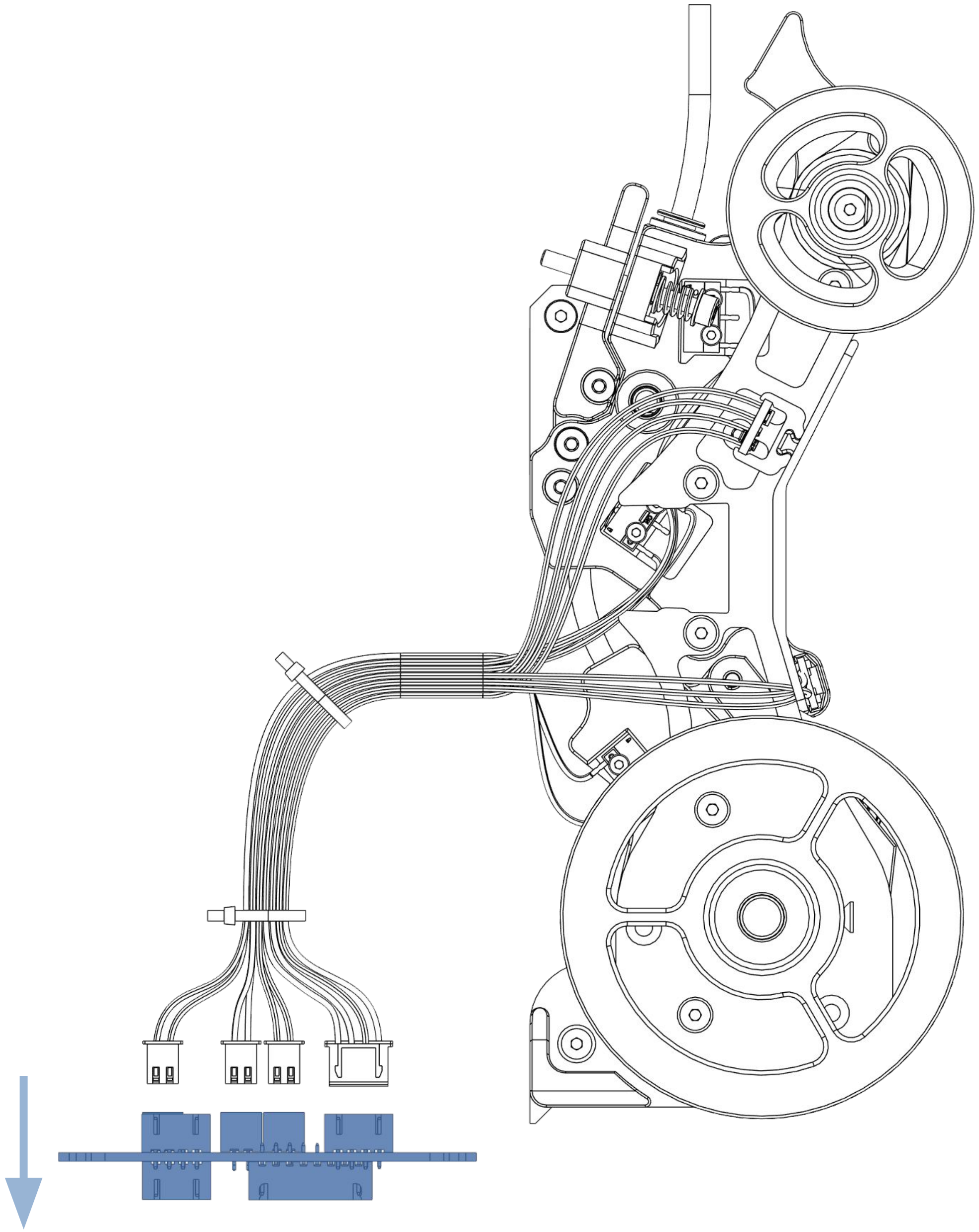
Stand the unit and hatch board as shown and add the cable ties.

One cable tie needs to be close to the hatch board and one towards the filamentalist assembly.

This will prevent the wires from getting caught on the wheels during operation.



DISCONNECT ALL THE WIRES FROM THE HATCH BOARD



EMU Github
https://github.com/DW-Tas/EMU

See my other projects
https://dwtas.net/

Join us on Discord !
https://discord.gg/WuxGVTzu5p